

NEXUS DATALENS

An End-to-End Automated Data Analytics Pipeline on AWS

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Abstract

Nexus DataLens is a cloud-native, end-to-end automated data analytics pipeline built on Amazon Web Services (AWS). The project addresses a common challenge faced by businesses and data professionals — transforming raw, unstructured CSV data into meaningful insights without manual processing or deep technical expertise. Nexus DataLens streamlines this process by enabling users to upload raw CSV files through a simple web interface. From there, the system autonomously executes a sequence of operations: ingestion into Amazon S3, cleaning via AWS Glue DataBrew, schema cataloging through AWS Glue Crawlers, and visualization using Amazon QuickSight or Power BI. The outcome is a scalable, secure, and cost-effective solution that accelerates data-to-decision workflows with minimal user effort. Whether for internal dashboards or client reporting, Nexus DataLens demonstrates the power of serverless, event-driven architectures in modern data engineering.

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1. Introduction

1.1 Background

In today's data-driven environment, organizations collect large volumes of raw data from various sources. However, transforming this data into actionable insights typically requires technical expertise, manual cleaning, and lengthy processes. There is a growing need for scalable, automated solutions that can streamline data ingestion, processing, and visualization particularly for non-technical users. Nexus DataLens addresses this gap by providing a fully automated pipeline built on reliable AWS services.

1.2 Motivation

This project is motivated by the need to simplify and democratize data analytics. Manual workflows often introduce delays, errors, and scalability issues. With increasing data volumes, it becomes essential to implement a system that automates every step—from file ingestion to dashboard generation—thereby saving time, reducing effort, and enhancing accessibility for all users, regardless of technical background.

1.3 Scope of the Project

This project focuses on building an end-to-end data analytics solution capable of:

- Ingest CSV files via a web-based frontend
- Process and clean data automatically using AWS Glue and DataBrew
- Create and update Athena tables for SQL querying
- Visualize data dynamically in Amazon QuickSight and optionally Power BI
- Enables seamless, near real-time insights without human intervention

The scope is limited to CSV-based structured data and primarily uses AWS services under the free tier wherever possible.

1.4 Technologies Used

The Nexus DataLens project integrates a diverse range of AWS services and technologies to deliver a seamless, automated data analytics workflow. The technologies used are grouped below into **Frontend**, **Backend**, and **Support Services** for clarity:

Frontend Technologies

- **HTML, CSS, JavaScript** – Developed a clean, responsive user interface for CSV upload
- **Amazon S3 (Static Website Hosting)** – Hosts the frontend as a static web application
- **AWS Cognito** – Provides secure user authentication
- **Upload Path** – Files are uploaded directly to:
s3://nexus-web-frontend1/data/filename.csv

Backend & Data Processing Technologies

- **Amazon S3** – Stores both raw uploaded files and cleaned outputs
- **AWS Lambda** – Automates Glue Crawlers and Glue Jobs upon file upload
- **AWS Glue Crawler** – Detects schema from new data files and updates Glue Catalog
- **AWS Glue (Spark-based Jobs)** – Cleans and transforms CSV data efficiently
- **AWS Glue DataBrew** – Provides no-code UI-based transformation for advanced cleaning
- **Amazon Athena** – Performs SQL queries on cleaned datasets stored in S3
- **Amazon QuickSight** – Creates dynamic, interactive dashboards from Athena datasets
- **Power BI (optional)** – Enterprise-level reporting via Athena ODBC integration
- **Python** – Used in AWS Glue Jobs and Lambda scripts for logic implementation

Security, Monitoring & Integration Services

- **AWS IAM** – Role-based access control for services and users
- **AWS KMS** – Key management for data encryption at rest
- **AWS CloudWatch** – Real-time logging and monitoring of Lambda and Glue operations
- **AWS API Gateway** – Exposes RESTful endpoints if integrating external tools
- **Amazon CloudFront** – Content delivery network (CDN) for global access to the frontend
- **AWS Cognito** – Adds user pool-based authentication to restrict uploads

4. Problem Statement

In many organizations, data analysts and business users often spend significant time manually processing and cleaning uploaded data for reporting purposes. This traditional approach is not only time-consuming but also prone to human error, lacks scalability, and delays decision-making. There is a growing need for an automated mechanism that can reliably process raw user-uploaded CSV files and generate meaningful insights with minimal technical effort.

5. Solution Overview

Nexus DataLens addresses this challenge by implementing a fully automated, event-driven data analytics pipeline built on AWS. The architecture eliminates manual intervention and delivers real-time insights using the following key services:

- **Amazon S3** – for secure and scalable file storage
- **AWS Lambda** – for automating workflow execution upon file upload
- **AWS Glue (Crawler + Job)** – to infer schema and perform data cleaning and transformation
- **Amazon Athena** – to query structured, cleaned datasets using standard SQL
- **Amazon QuickSight** (*and optionally Microsoft Power BI*) – to build rich, interactive dashboards for end users

The entire pipeline is cloud-native, scalable, and designed for seamless integration, ensuring data moves from ingestion to visualization without any manual processing.

6. Objectives

The primary objectives of the Nexus DataLens project are as follows:

- To build a robust and scalable platform using AWS services for automated CSV data ingestion
- To clean and transform datasets using AWS Glue with no manual effort
- To enable seamless data visualization through **Amazon QuickSight** and **Power BI**
- To create a frontend user interface that allows non-technical users to upload datasets and receive insights effortlessly

Note: This project supports visualization through both Amazon QuickSight and Microsoft Power BI.

Both platforms were successfully integrated and tested during implementation.

Architecture Diagram

Thanks for sharing the updated architecture! Here's how you can **integrate this diagram and description** into your final project document under the **Architecture** section:

3.5 Architecture Overview

The architecture of **Nexus DataLens** is designed for **end-to-end automation**, **real-time data processing**, and **scalable analytics** — all built on a **serverless AWS stack**.

System Architecture Diagram

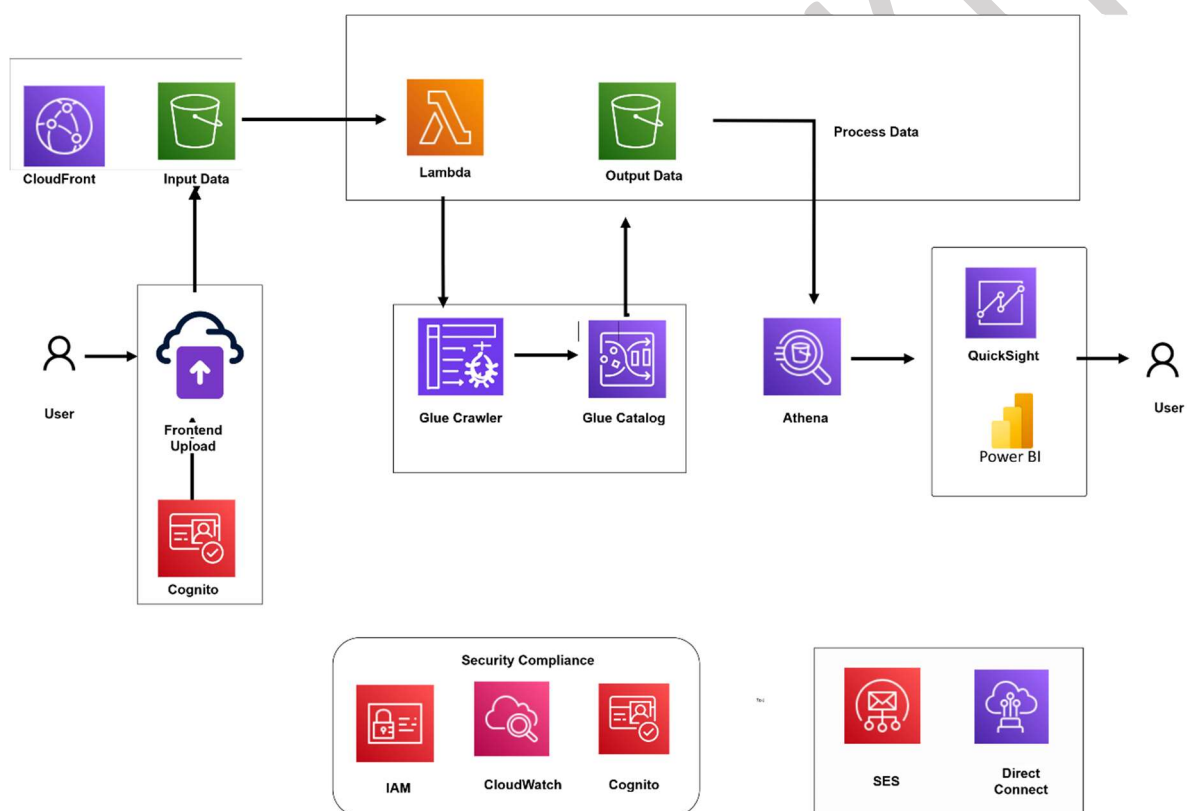


Figure 1: Nexus DataLens – End-to-End Serverless Architecture

Data Flow Explanation

This architecture enables seamless data ingestion, processing, and visualization using AWS services. Here's how the components interact:

1. Frontend Upload & Authentication

- Users upload CSV files through a web-based frontend.
- **Amazon Cognito** secures access with optional user authentication.
- Files are stored in the S3 input folder and served via **CloudFront**.

2. AWS Lambda Orchestration

- The S3 upload triggers **AWS Lambda**, which initiates:
 - A **Glue Crawler** to scan and update the **Glue Catalog**
 - A **Glue ETL Job** to clean and format the data
- Cleaned data is written to the cleaned-output/ folder in S3.

3. Athena & Data Querying

- Cleaned data is queried using **Amazon Athena**.
- Athena integrates with both **QuickSight** and **Power BI** for dashboarding.

4. Visualization & Reporting

- Users can visualize the data using **Amazon QuickSight** or **Microsoft Power BI** dashboards.

5. Security & Compliance

- **IAM** ensures access control.
- **CloudWatch** handles monitoring and alerts.
- **Cognito** adds secure authentication for both the frontend and backend services.

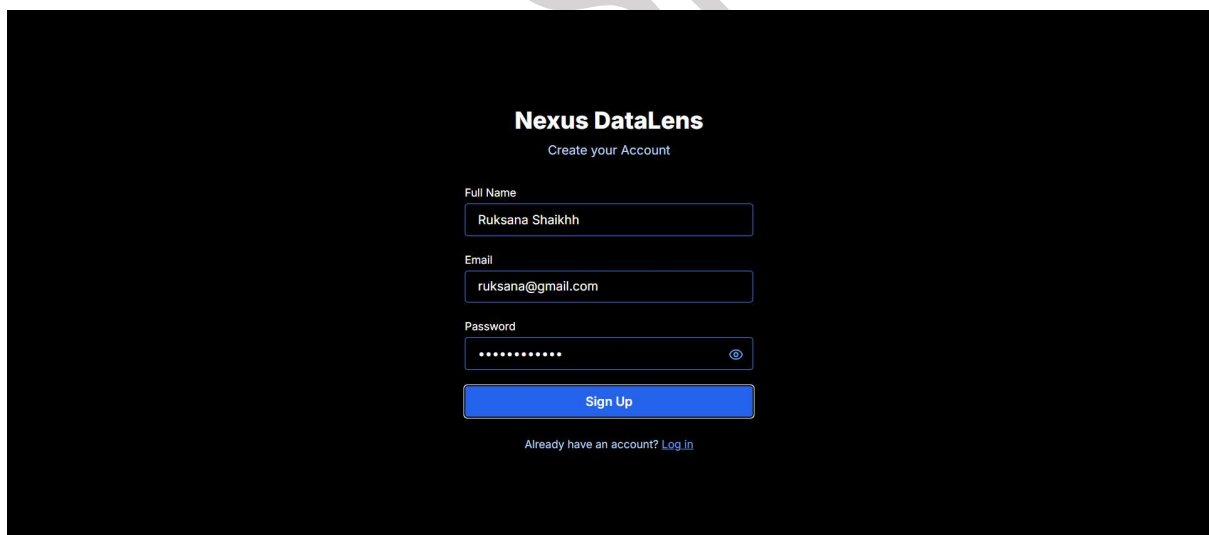
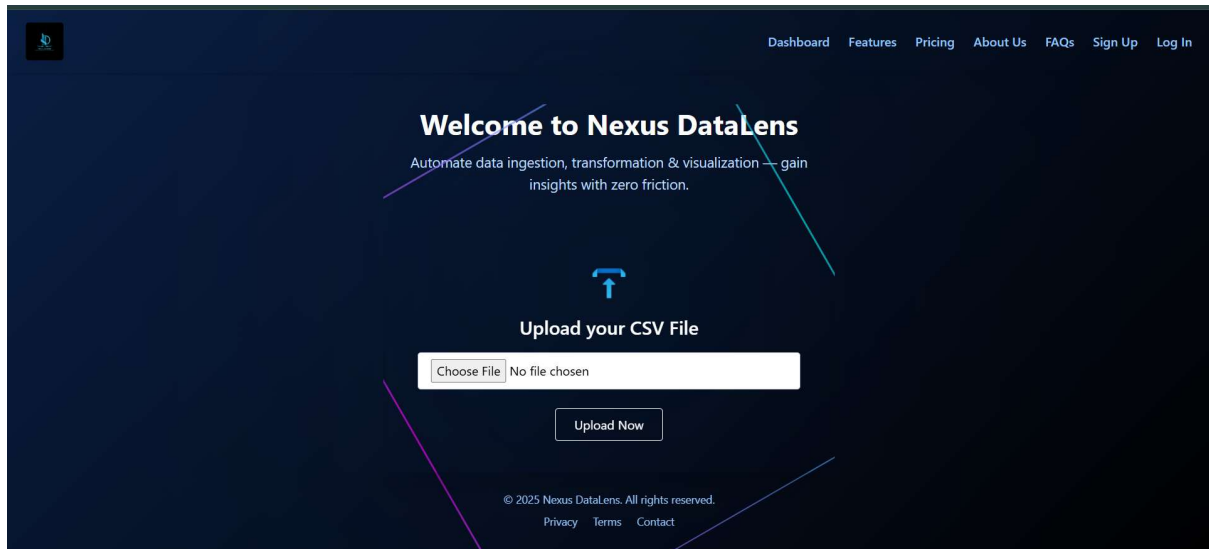
6. Optional Enterprise Features

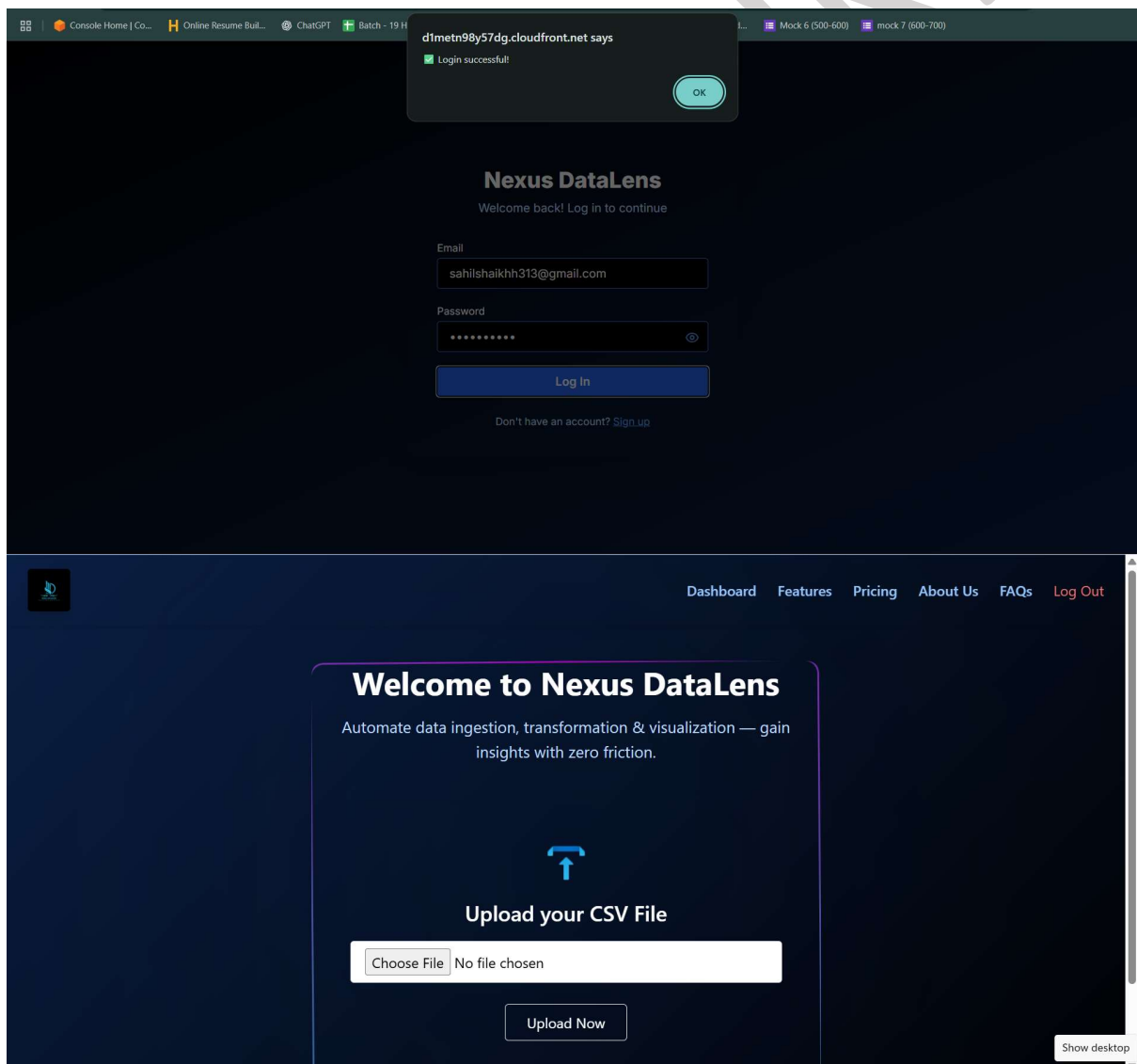
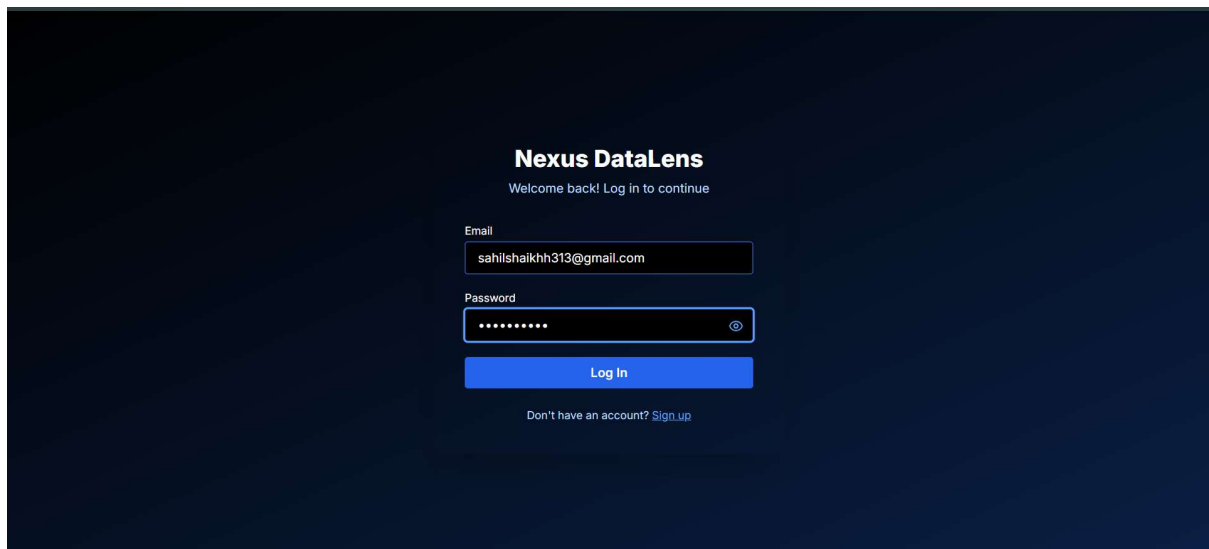
- **Amazon SES** can send automated email reports/alerts.
- **AWS Direct Connect** can be used in enterprise setups for faster, private network access.

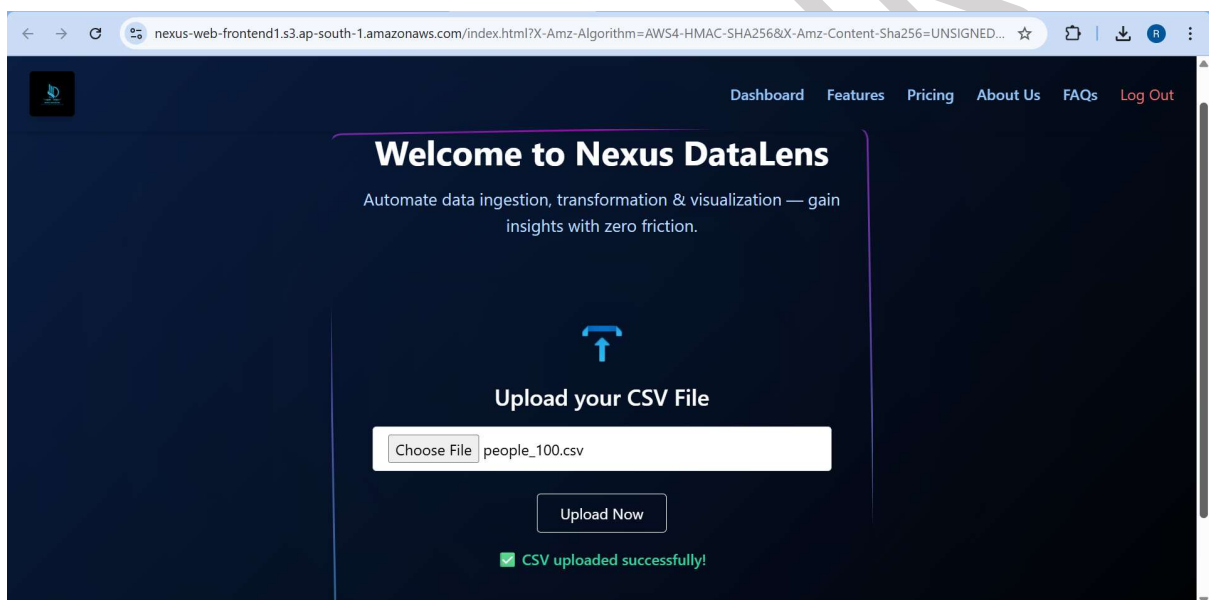
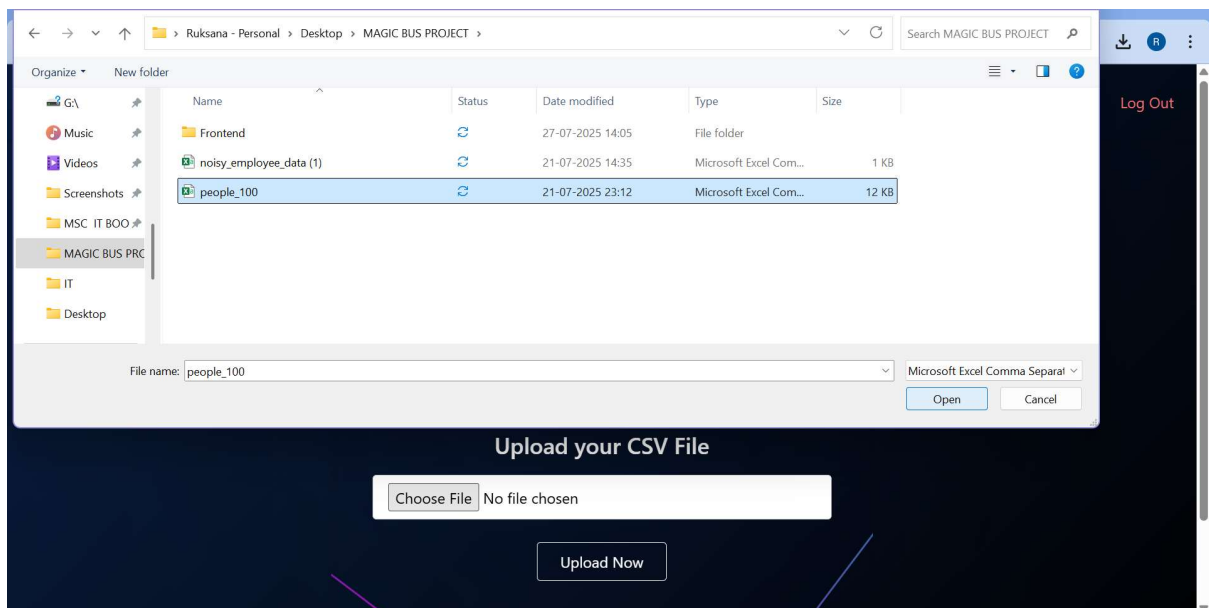
5. Frontend Application

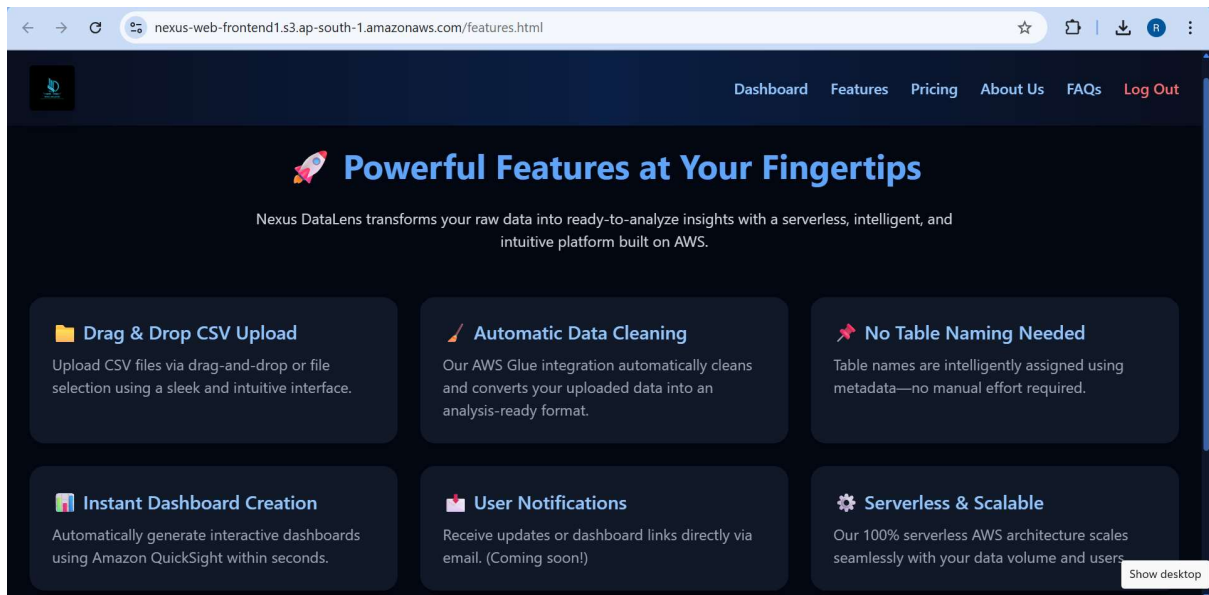
5.1 Upload Interface

This section covers the CSV file upload mechanism built using a simple HTML/CSS/JavaScript interface hosted on Amazon S3.





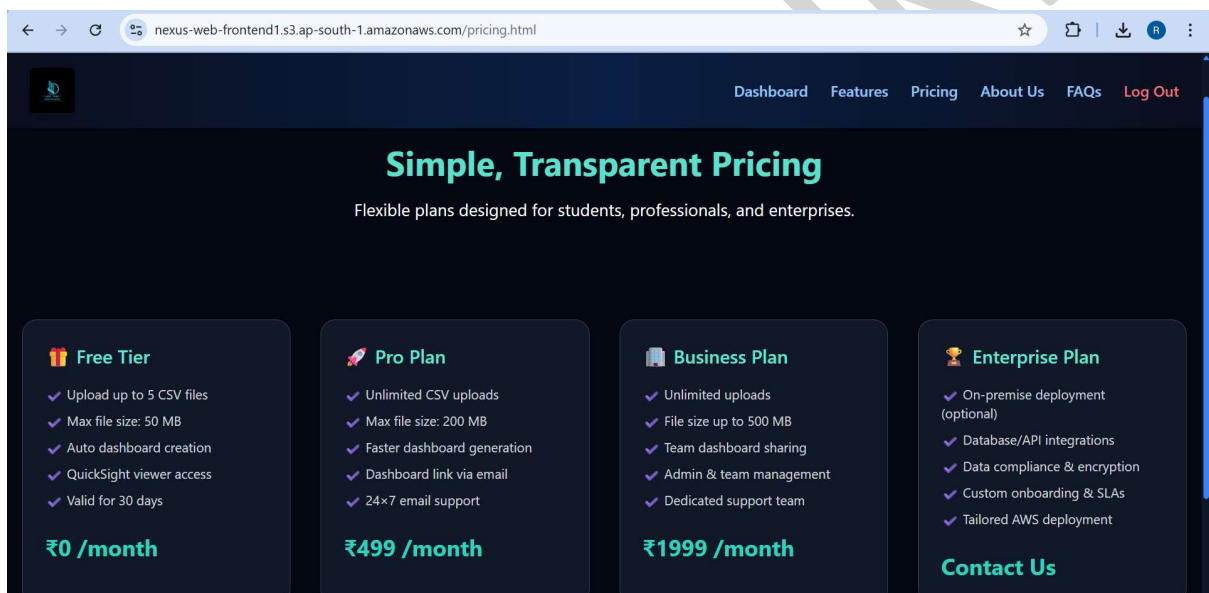




The screenshot shows the 'Features' page of the Nexus DataLens web application. The browser address bar displays 'nexus-web-frontend1.s3.ap-south-1.amazonaws.com/features.html'. The navigation bar includes links for Dashboard, Features, Pricing, About Us, FAQs, and Log Out. The main heading is 'Powerful Features at Your Fingertips', followed by a subtext: 'Nexus DataLens transforms your raw data into ready-to-analyze insights with a serverless, intelligent, and intuitive platform built on AWS.' Below this, there are six feature cards arranged in a 2x3 grid:

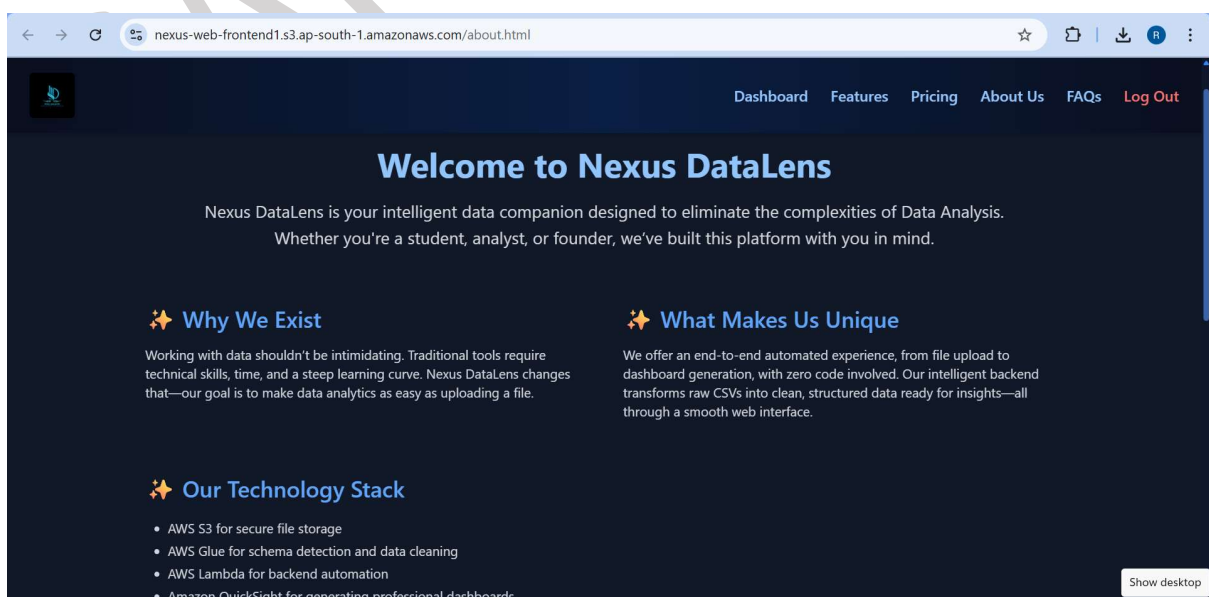
- Drag & Drop CSV Upload**: Upload CSV files via drag-and-drop or file selection using a sleek and intuitive interface.
- Automatic Data Cleaning**: Our AWS Glue integration automatically cleans and converts your uploaded data into an analysis-ready format.
- No Table Naming Needed**: Table names are intelligently assigned using metadata—no manual effort required.
- Instant Dashboard Creation**: Automatically generate interactive dashboards using Amazon QuickSight within seconds.
- User Notifications**: Receive updates or dashboard links directly via email. (Coming soon!)
- Serverless & Scalable**: Our 100% serverless AWS architecture scales seamlessly with your data volume and users.

A 'Show desktop' button is visible in the bottom right corner.



The screenshot shows the 'Pricing' page of the Nexus DataLens web application. The browser address bar displays 'nexus-web-frontend1.s3.ap-south-1.amazonaws.com/pricing.html'. The navigation bar includes links for Dashboard, Features, Pricing, About Us, FAQs, and Log Out. The main heading is 'Simple, Transparent Pricing', followed by a subtext: 'Flexible plans designed for students, professionals, and enterprises.' Below this, there are four pricing plans arranged in a row:

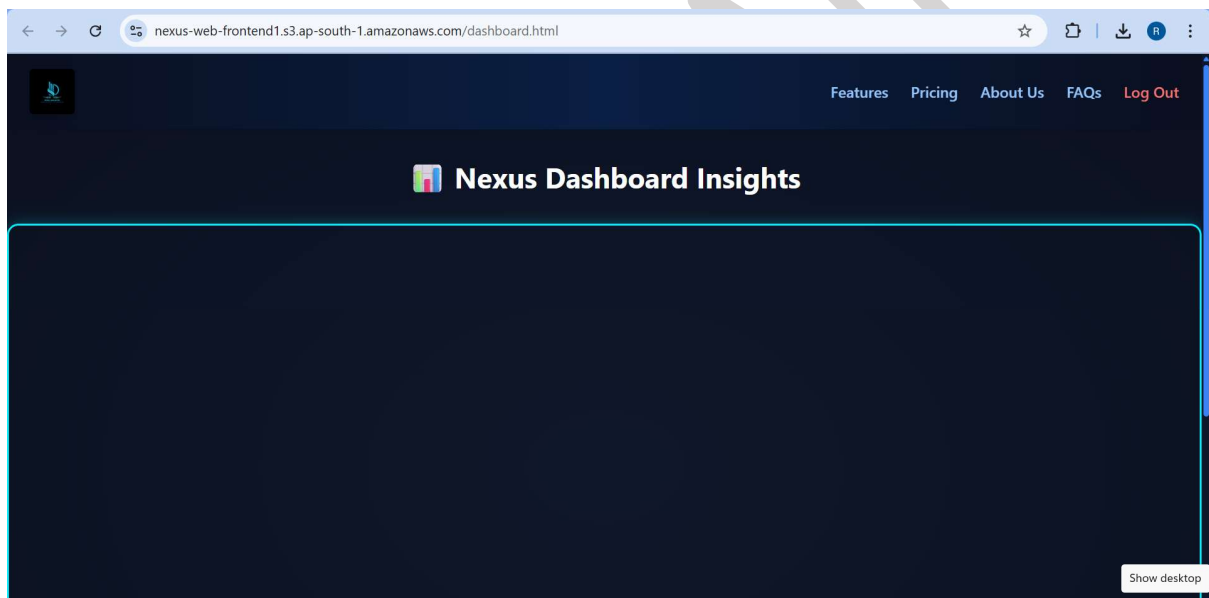
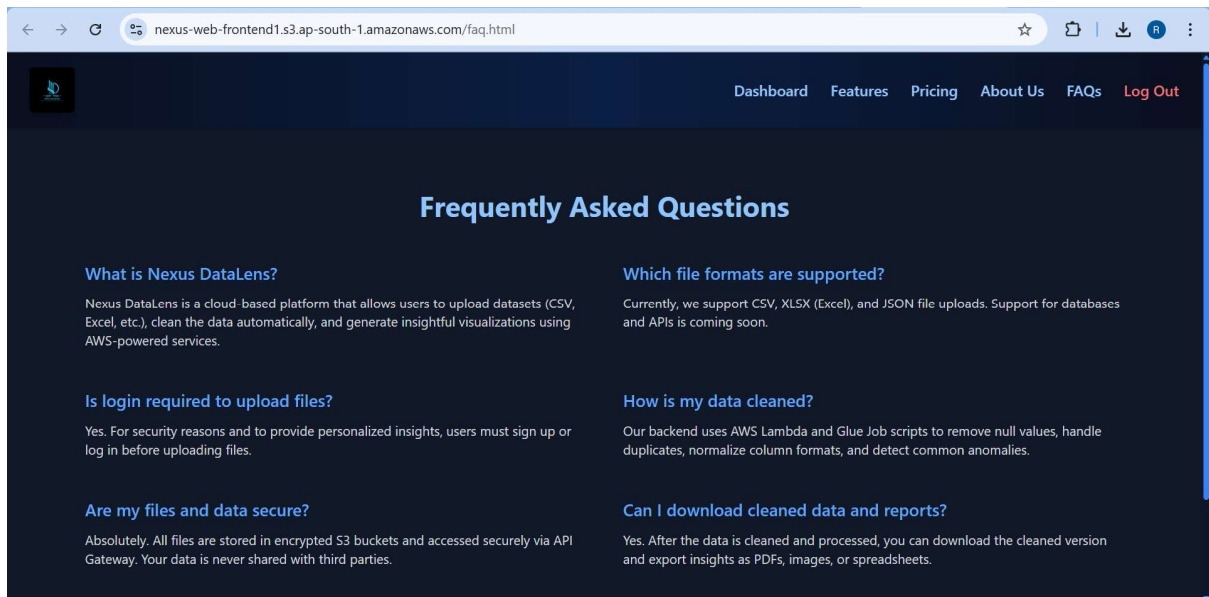
- Free Tier**: Upload up to 5 CSV files, Max file size: 50 MB, Auto dashboard creation, QuickSight viewer access, Valid for 30 days. Price: ₹0 /month.
- Pro Plan**: Unlimited CSV uploads, Max file size: 200 MB, Faster dashboard generation, Dashboard link via email, 24x7 email support. Price: ₹499 /month.
- Business Plan**: Unlimited uploads, File size up to 500 MB, Team dashboard sharing, Admin & team management, Dedicated support team. Price: ₹1999 /month.
- Enterprise Plan**: On-premise deployment (optional), Database/API integrations, Data compliance & encryption, Custom onboarding & SLAs, Tailored AWS deployment. Price: Contact Us.

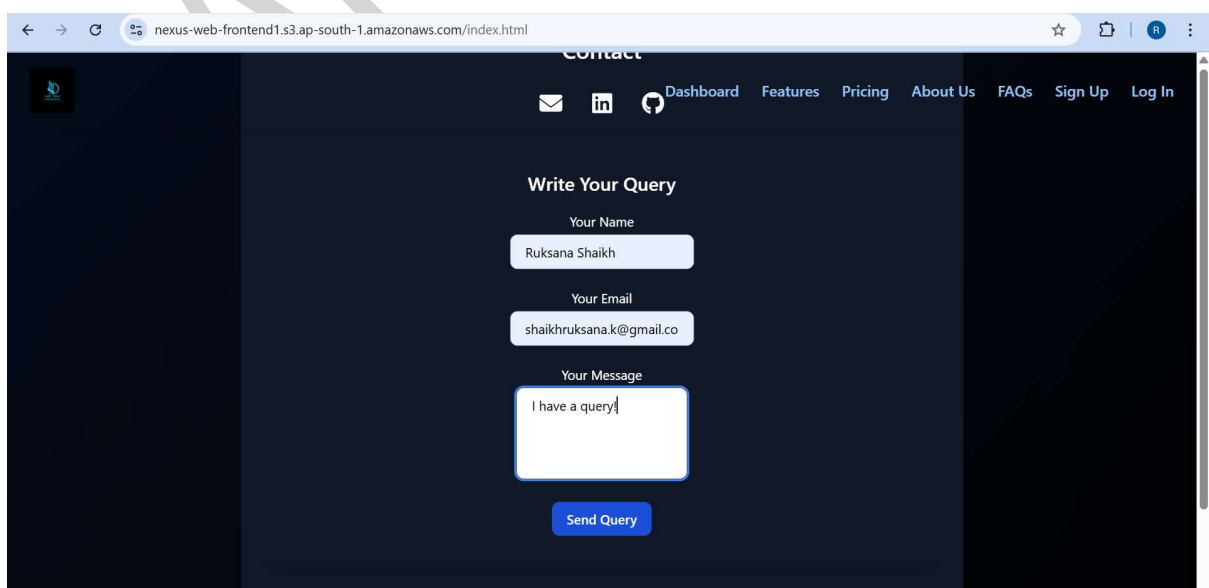
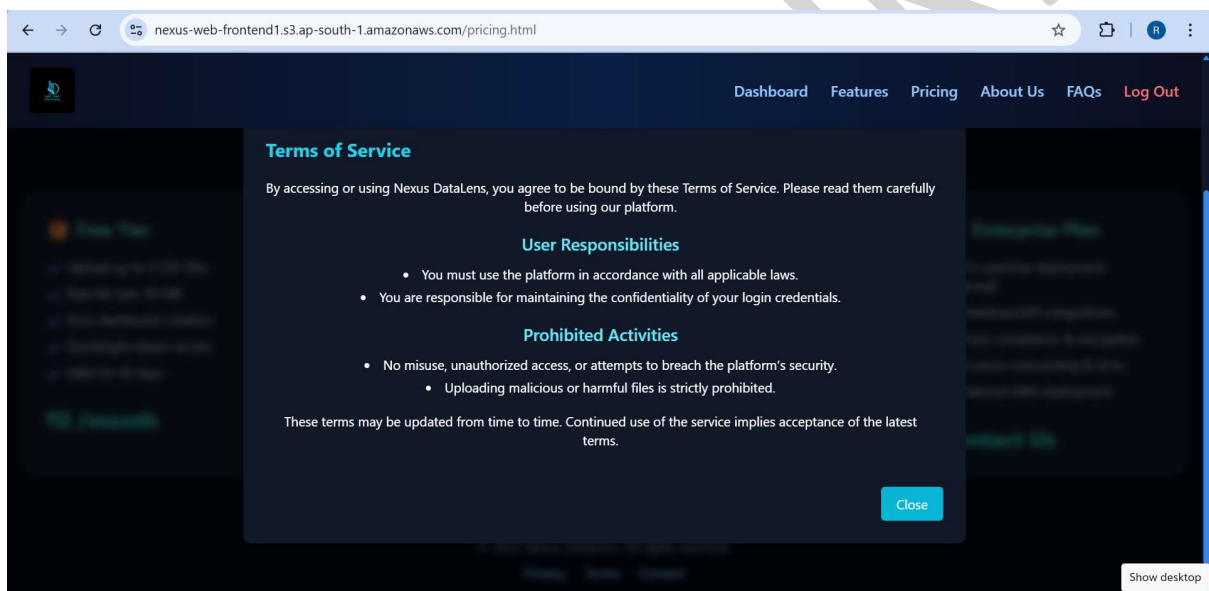
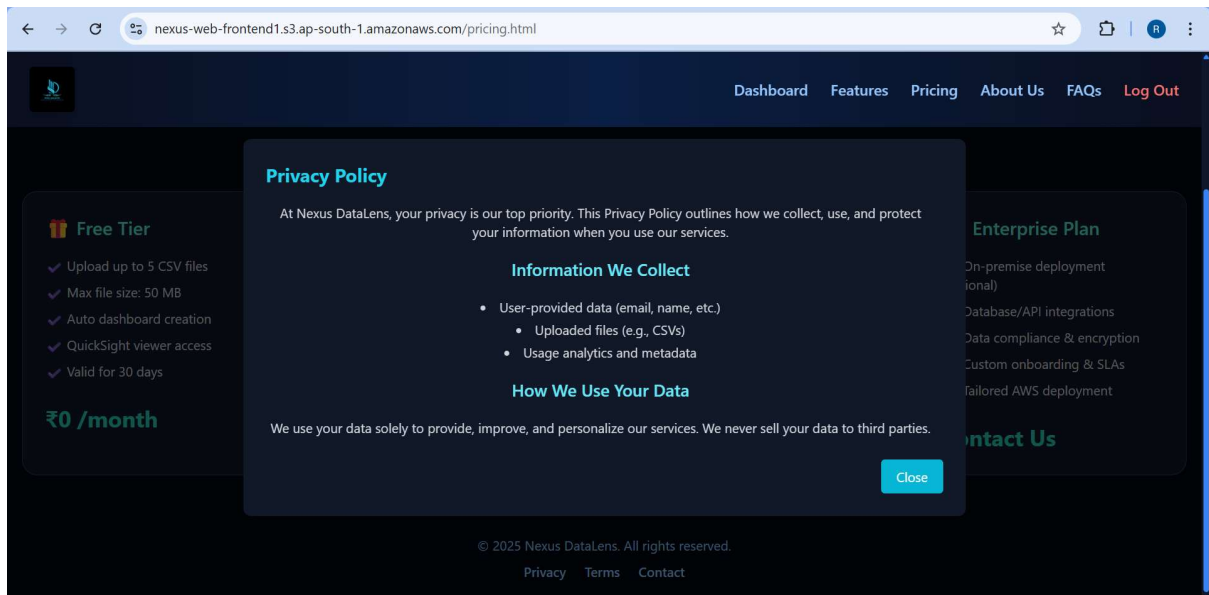


The screenshot shows the 'About' page of the Nexus DataLens web application. The browser address bar displays 'nexus-web-frontend1.s3.ap-south-1.amazonaws.com/about.html'. The navigation bar includes links for Dashboard, Features, Pricing, About Us, FAQs, and Log Out. The main heading is 'Welcome to Nexus DataLens', followed by a subtext: 'Nexus DataLens is your intelligent data companion designed to eliminate the complexities of Data Analysis. Whether you're a student, analyst, or founder, we've built this platform with you in mind.' Below this, there are three sections:

- Why We Exist**: Working with data shouldn't be intimidating. Traditional tools require technical skills, time, and a steep learning curve. Nexus DataLens changes that—our goal is to make data analytics as easy as uploading a file.
- What Makes Us Unique**: We offer an end-to-end automated experience, from file upload to dashboard generation, with zero code involved. Our intelligent backend transforms raw CSVs into clean, structured data ready for insights—all through a smooth web interface.
- Our Technology Stack**:
 - AWS S3 for secure file storage
 - AWS Glue for schema detection and data cleaning
 - AWS Lambda for backend automation
 - Amazon QuickSight for generating professional dashboards

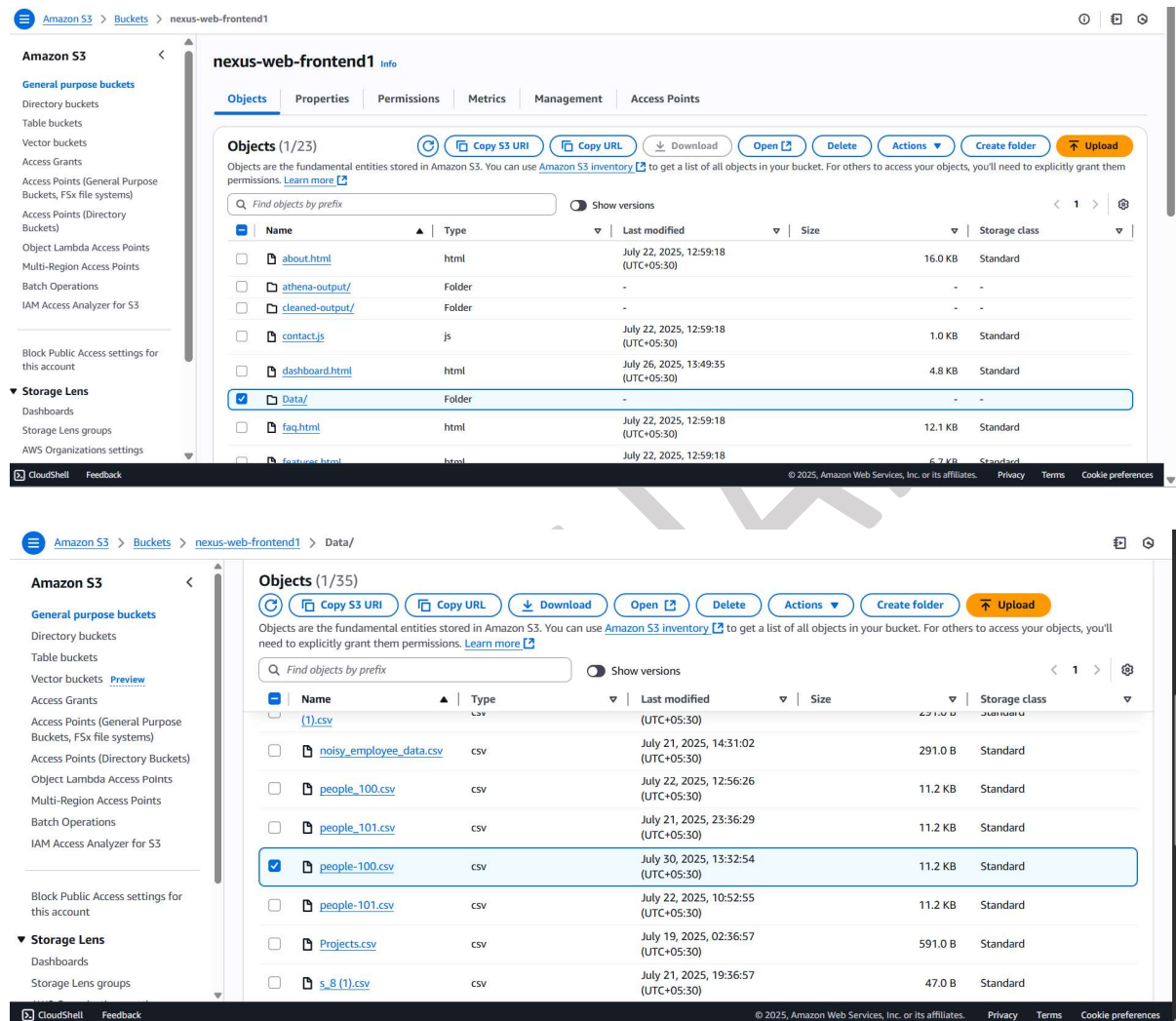
A 'Show desktop' button is visible in the bottom right corner.





5.2 S3 Bucket Structure

Overview of the S3 bucket layout including folders for raw and cleaned data.



4. Project Workflow – Step-by-Step Breakdown

The **Nexus DataLens** project leverages a robust serverless architecture to automate the entire data lifecycle — from file upload to advanced business visualizations. Below is a comprehensive breakdown of how each AWS service integrates into the workflow:

Step 1: Frontend File Upload and Hosting

The frontend is a **React-based web application**, hosted on **Amazon S3 (Static Website Hosting)** where users interact to upload .csv files.

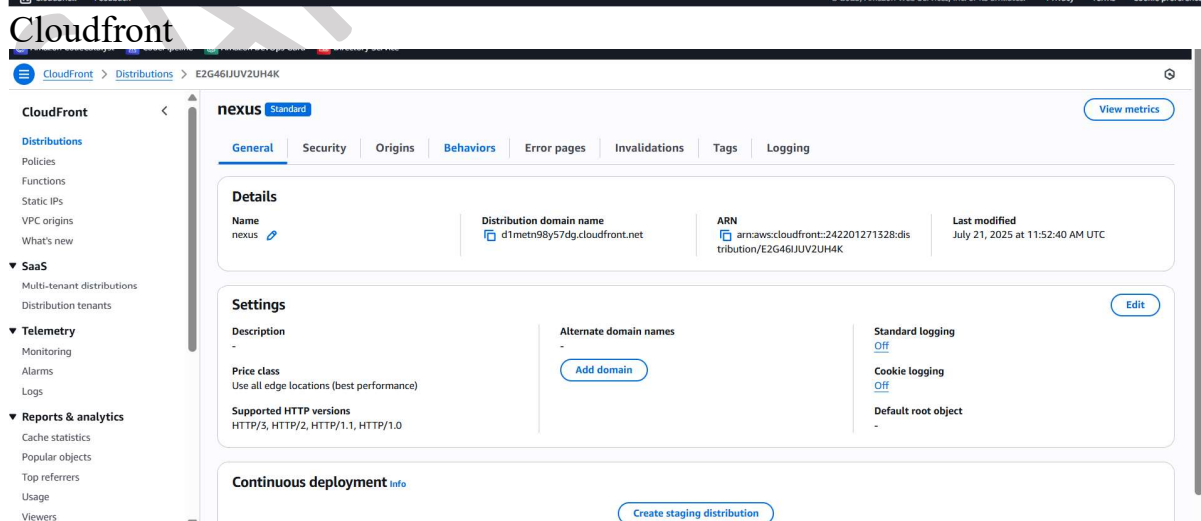
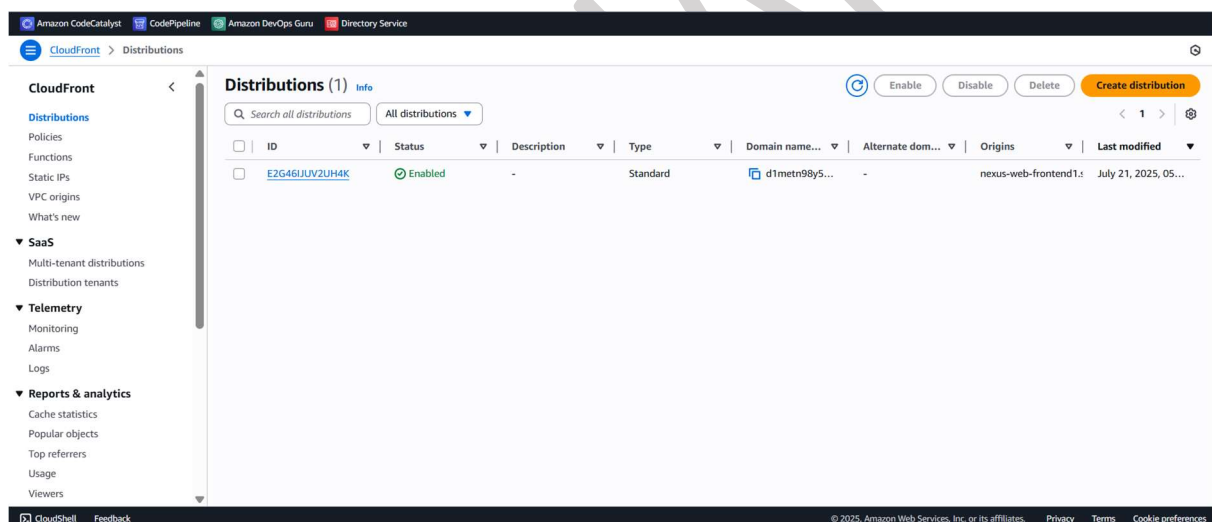
- Uploaded files are securely stored in:
- `s3://nexus-web-frontend1/data/`
- **Amazon CloudFront** is used as a CDN to deliver the frontend globally with low latency.
- **AWS Cognito** enables **secure user authentication**, ensuring that only authorized users can upload CSV files.

✂ Technologies Used:

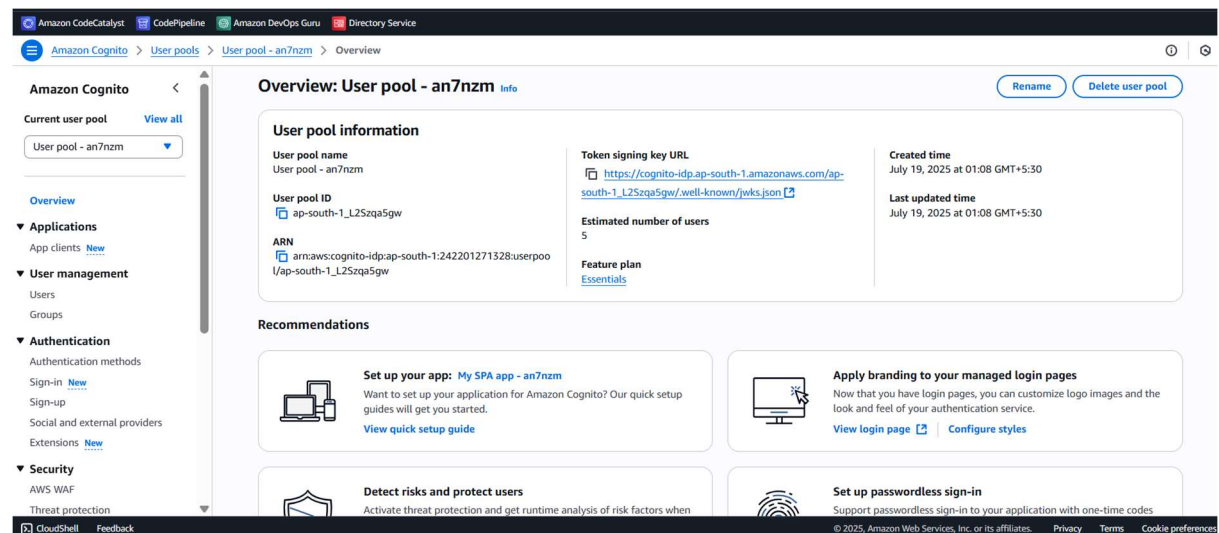
- Amazon S3 (Website Hosting)
- Amazon CloudFront (CDN)
- AWS Cognito (Authentication)

📷 Suggested Screenshots:

- Frontend UI with file upload section
- S3 bucket hosting the static site (console)
- Cognito user pool dashboard



Cloudfront



cognito

Step 2: Secure File Storage in Amazon S3

- Upon upload, CSV files are stored in the raw data folder:

bash

Copy code

s3://nexus-web-frontend1/data/

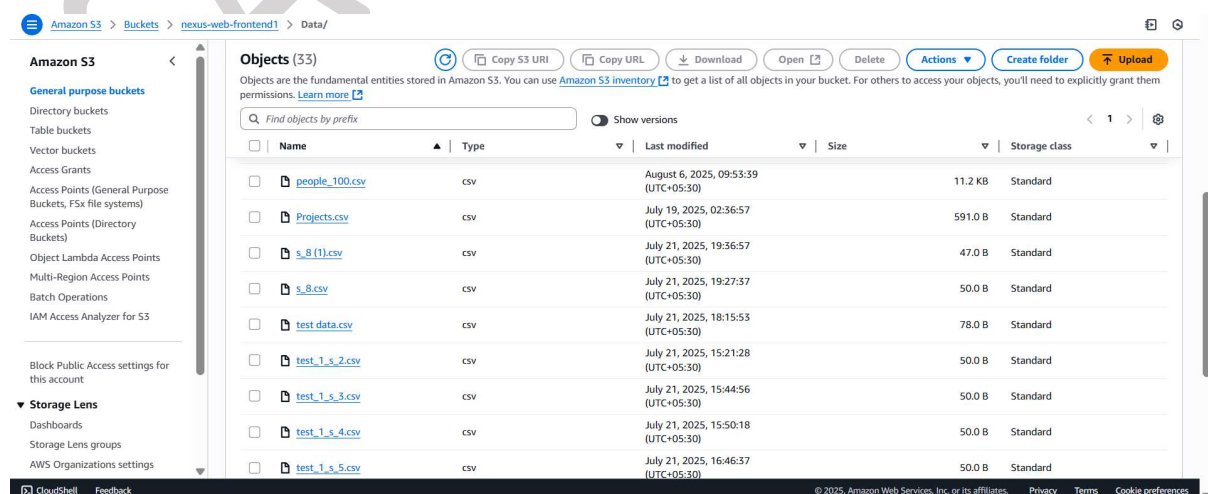
- AWS KMS (Key Management Service)** ensures encryption at rest.
- Fine-grained access is controlled using **AWS IAM roles and policies**.

✂ Technologies Used:

- Amazon S3 (Data Storage)
- AWS IAM (Access Control)
- AWS KMS (Encryption)

📷 Suggested Screenshots:

- S3 bucket structure with uploaded CSV in data/
- IAM role used for S3 access
- KMS key policy for encryption



Data Upload BUCKET

Summary

Creation date: July 19, 2025, 10:14 (UTC+05:30)

Last activity: 16 minutes ago

ARN: arn:aws:iam::242201271328:role/NEXUS_USER_ROLE

Maximum session duration: 1 hour

Permissions policies (4)

Policy name	Type	Attached entities
AmazonCognitoPowerUser	AWS managed	1
AmazonDynamoDBFullAccess	AWS managed	2
AmazonS3FullAccess	AWS managed	English (India) English (India)
AWSLambdaBasicExecutionRole	AWS managed	

IAM Role

General configuration

Alias: alias/threatray-key

ARN: arn:aws:kms:ap-south-1:242201271328:key/0772dc11-a8b6-48cf-8a6e-b6109c991d43

Current key material ID: 9c72fbfbd5dc7ba4ffbb140618dc8f3f870a423e32aafa36fc409bcc7688cbd

Status: Enabled

Creation date: Jul 30, 2025 19:59 GMT+5:30

Regionality: Single Region

Key policy

```

1 {
2   "Version": "2012-10-17",
3   "Id": "Key policy created by CloudTrail",
4   "Statement": [
5     {
6       "Sid": "Enable IAM User Permissions",
7       "Effect": "Allow",

```

KMS

Step 3: Lambda Trigger for Automation

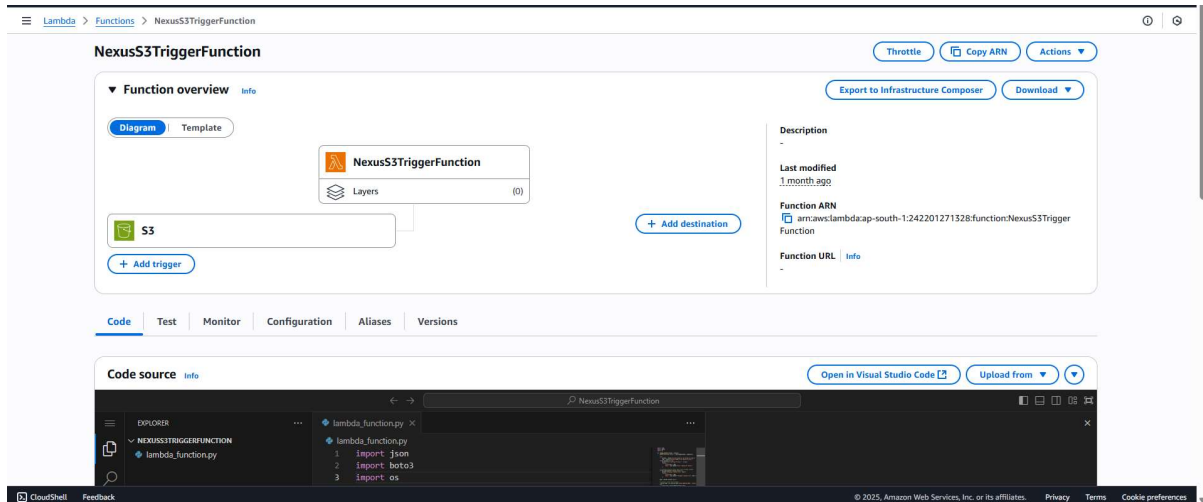
- An S3 PUT event notification automatically triggers the LambdaTriggerQuicksight Lambda function. It performs the following operations:
 - Invokes **Glue Crawler** to scan the raw file
 - Waits for crawler completion via polling logic
 - Runs the **Glue Job** for cleaning

Technologies Used:

- AWS Lambda
- Amazon S3 Event Notifications
- AWS CloudWatch (Monitoring)

Suggested Screenshots:

- Lambda trigger config (S3 event → Lambda)
- Lambda code snippet (Python)
- CloudWatch logs showing process



NexusS3TriggerFunction

Function overview

Diagram: S3 → NexusS3TriggerFunction

Description:

- Last modified: 1 month ago
- Function ARN: arn:aws:lambda:ap-south-1:242201271328:function:NexusS3TriggerFunction
- Function URL: info

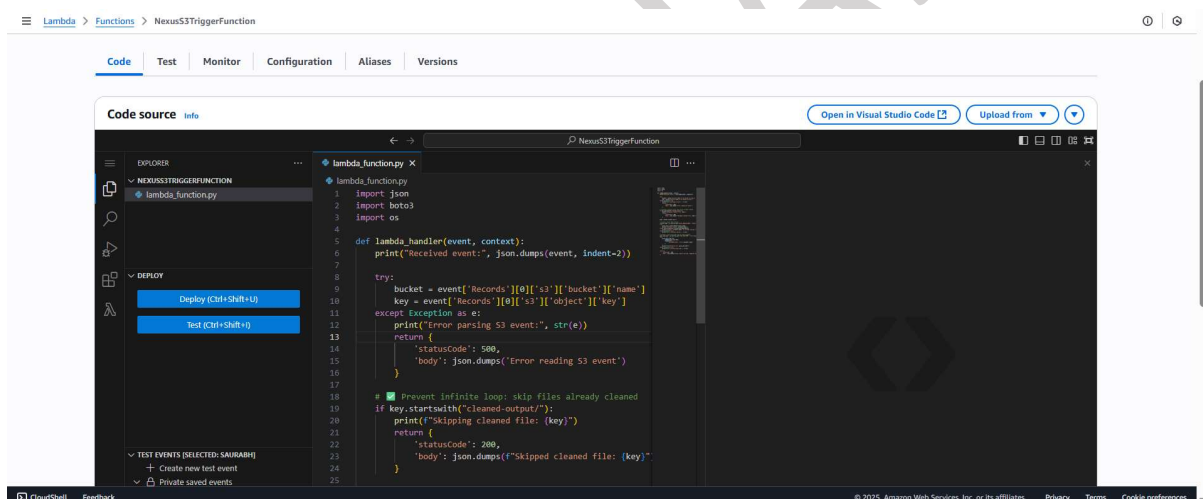
Code source

```

1 import json
2 import boto3
3 import os

4
5 def lambda_handler(event, context):
6     print("Received event:", json.dumps(event, indent=2))
7
8     try:
9         bucket = event['Records'][0]['s3']['bucket']['name']
10        key = event['Records'][0]['s3']['object']['key']
11    except Exception as e:
12        print("Error parsing S3 event:", str(e))
13        return {
14            'statusCode': 500,
15            'body': json.dumps('Error reading S3 event')
16        }
17
18    # Prevent infinite loop: skip files already cleaned
19    if key.startswith("cleaned-output/"):
20        print("Skipping cleaned file: (key)")
21        return {
22            'statusCode': 200,
23            'body': json.dumps("Skipped cleaned file: (key)")
24        }
25
26

```



Code source

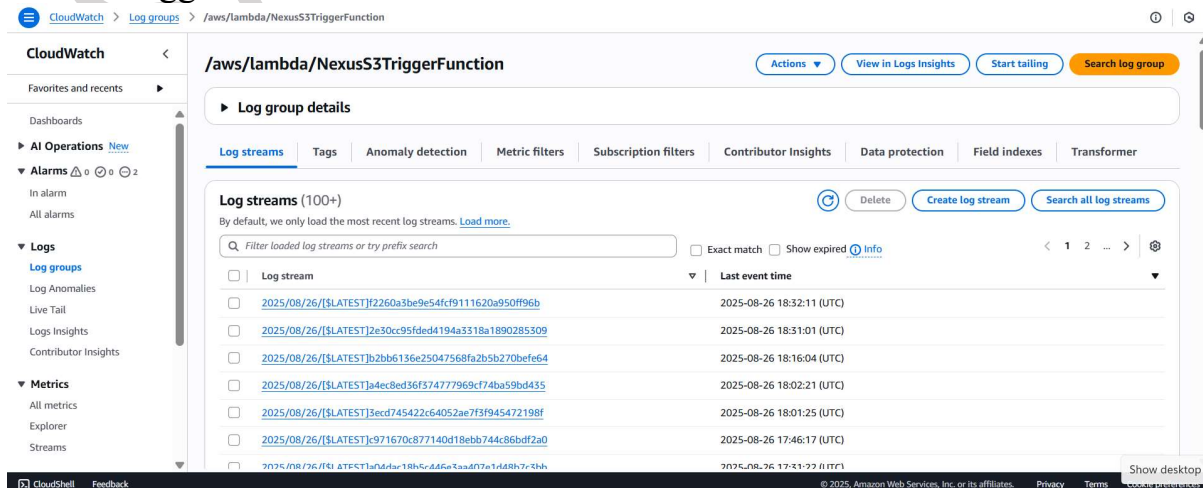
```

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11    except Exception as e:
12        print("Error parsing S3 event:", str(e))
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14            'statusCode': 500,
15            'body': json.dumps('Error reading S3 event')
16        }
17
18    # Prevent infinite loop: skip files already cleaned
19    if key.startswith("cleaned-output/"):
20        print("Skipping cleaned file: (key)")
21        return {
22            'statusCode': 200,
23            'body': json.dumps("Skipped cleaned file: (key)")
24        }
25
26

```

lambda trigger function



CloudWatch

/aws/lambda/NexusS3TriggerFunction

Log group details

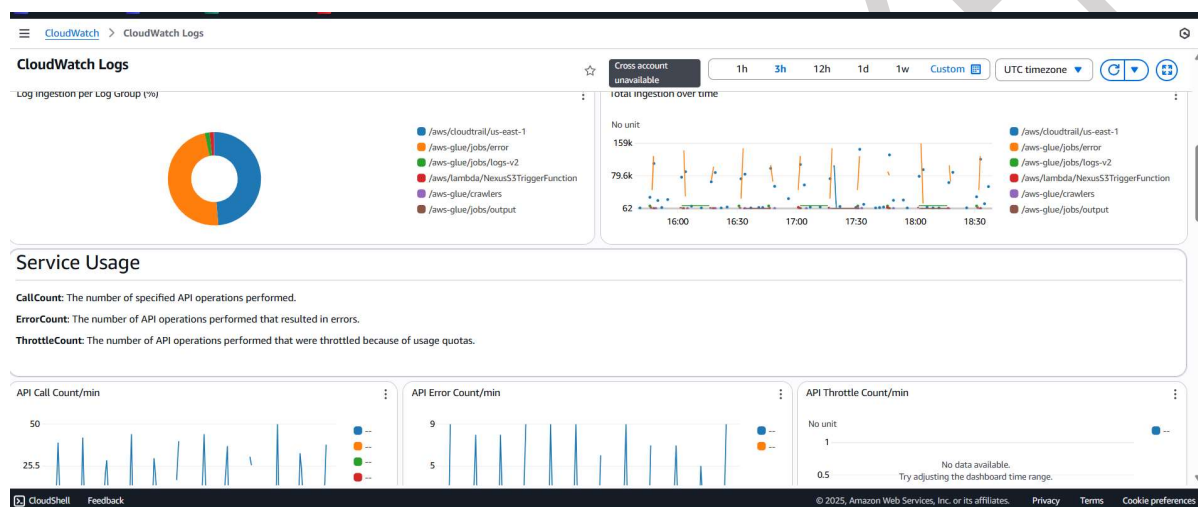
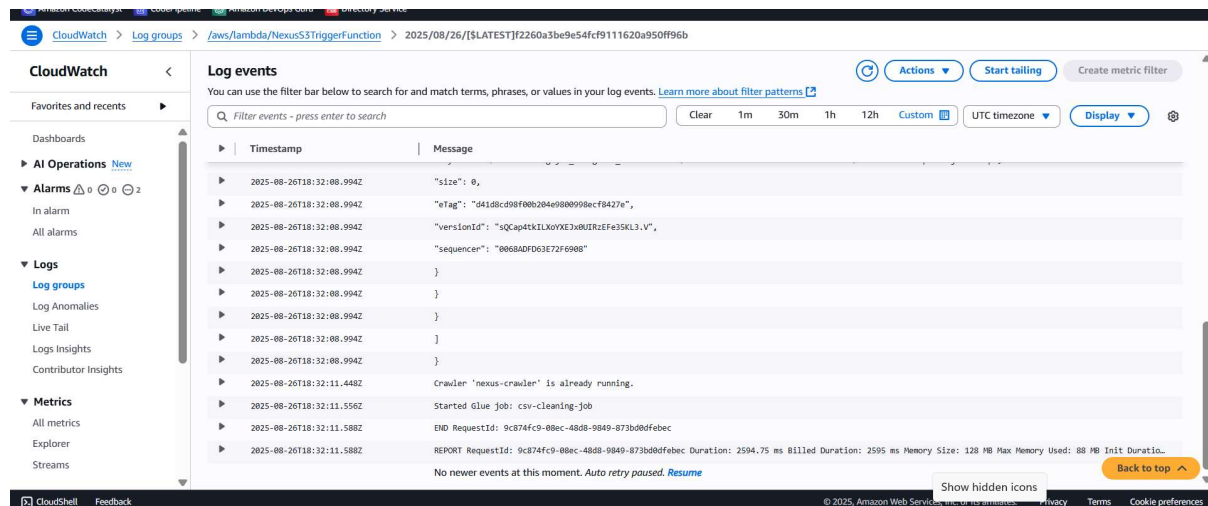
Log streams (100+)

By default, we only load the most recent log streams. [Load more](#)

Filter loaded log streams or try prefix search

Exact match Show expired Info

Log stream	Last event time
2025/08/26/[{"\$LATEST"}]f2260a3be9e54fcf9111620a950ff96b	2025-08-26 18:32:11 (UTC)
2025/08/26/[{"\$LATEST"}]2e30cc95fde4194a3318a1890285309	2025-08-26 18:31:01 (UTC)
2025/08/26/[{"\$LATEST"}]b2bb6136e25047568fa2b5b270befe64	2025-08-26 18:16:04 (UTC)
2025/08/26/[{"\$LATEST"}]a4ec8ed36f374777969cf74ba59bd435	2025-08-26 18:02:21 (UTC)
2025/08/26/[{"\$LATEST"}]3ecd745422c64052ae7f3f945472198f	2025-08-26 18:01:25 (UTC)
2025/08/26/[{"\$LATEST"}]c971670c877140d18ebb744c86bdf2a0	2025-08-26 17:46:17 (UTC)
2025/08/26/[{"\$LATEST"}]a4d4a18f6c4d6e3aad7e14488b7c3b4	2025-08-26 17:41:22 (UTC)



cloudwatch

Step 4: Schema Detection with AWS Glue Crawler

- The Glue Crawler scans the uploaded file in the data/ folder, detects the schema, and updates the **AWS Glue Data Catalog**.
- The resulting table is used by **Athena** for SQL querying.

✂ Technologies Used:

- **AWS Glue Crawler**
- **AWS Glue Catalog**



Screenshot Suggestions:

- Glue Crawler configuration page
- Table in Glue Catalog pointing to data/ S3 path

Amazon CodeCatalyst

CodePipeline

Amazon DevOps Guru

Directory Service

AWS Glue

Crawlers

Getting started

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Data Catalog

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

Data Integration and ETL

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Interactive Sessions

Data classification tools

CloudShell

Feedback

Crawlers

Info

A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema for your data, and then creates metadata tables in your data catalog.

View and manage all available crawlers.

Filter crawlers

Name

State

Schedule

Last run

Last run timestamp

Log

Table changes from L...

cleaned-output-crawler

Ready

Succeeded

July 21, 2025 at 12:4...

View log

45 created

nexus-crawler

Ready

Succeeded

August 26, 2025 at 18...

View log

-

Last updated (UTC)

August 26, 2025 at 19:06:47

Action

Run

Create crawler

Amazon CodeCatalyst

CodePipeline

Amazon DevOps Guru

Directory Service

AWS Glue

Crawlers

nexus-crawler

Getting started

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Data Catalog

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

Data Integration and ETL

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Interactive Sessions

Data classification tools

Sensitive data detection

Record Matching

Triggers

CloudShell

Feedback

nexus-crawler

Info

Last updated (UTC)

August 26, 2025 at 19:06:55

Run crawler

Edit

Delete

Crawler properties

Name

nexus-crawler

IAM role

GlueDataBrewRole_1

Database

nexusdb

State

READY

Description

-

Security configuration

-

Lake Formation configuration

-

Table prefix

-

Maximum table threshold

-

Advanced settings

Crawler runs

Schedule

Data sources

Classifiers

Tags

Crawler runs (20)

Info

The list of crawler runs for this crawler.

Filter by a date and time range

Start time (UTC)

End time (UTC)

Current/last duration

Status

DPU hours

Table changes

August 26, 2025 at 18:46:02

August 26, 2025 at 18:46:58

55 s

Completed

0.064

-

August 26, 2025 at 18:31:03

August 26, 2025 at 18:32:30

01 min 26 s

Completed

0.073

-

August 26, 2025 at 18:16:03

August 26, 2025 at 18:17:24

01 min 20 s

Completed

0.071

-

August 26, 2025 at 18:01:24

August 26, 2025 at 18:03:17

01 min 52 s

Completed

0.045

-

August 26, 2025 at 17:46:17

August 26, 2025 at 17:47:23

01 min 06 s

Completed

0.072

-

Amazon CodeCatalyst

CodePipeline

Amazon DevOps Guru

Directory Service

AWS Glue

Tables

Getting started

ETL jobs

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Tables (1/33)

Info

View and manage all available tables.

Filter tables

Name

Database

Location

Classification

Deprecated

View data

Data quality

Column statis...

deliveries_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

heart_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

index_html

nexusdb

s3://nexus-web-fronter

UNKNOWN

-

Table data

View data quality

View statistics

industry_csv

nexusdb

s3://nexus-web-fronter

UNKNOWN

-

Table data

View data quality

View statistics

loan_prediction_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

manifest_json

nexusdb

s3://nexus-web-fronter

JSON

-

Table data

View data quality

View statistics

mobile_price_prediction

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

nexusfile_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

nexusfile_csv_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

noisy_employee_data_

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

noisy_employee_data_

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

people_100_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

projects_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

s_8_1_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

s_8_csv

nexusdb

s3://nexus-web-fronter

CSV

-

Table data

View data quality

View statistics

22

The top screenshot shows the AWS Glue console interface for a table named 'people_100_csv'. The 'Schema' tab is selected, showing a list of columns with their names, data types, and partition keys. The columns are: index (bigint), user id (string), first name (string), last name (string), sex (string), email (string), phone (string), date of birth (string), and job title (string). The bottom screenshot shows the AWS Glue Studio interface for the 'Jobs' page. It displays a list of jobs, including 'csv-cleaning-job', which is a Glue ETL job created by a script on 8/13/2025 at 11:20:46 AM, using AWS Glue version 5.0.

Step 4: Glue Job – Data Cleaning

The Spark-based job csv-cleaning-job processes the raw data:

- Drops rows and columns that are entirely null
- Filters out invalid values
- Standardizes column headers
- Repartitions to a single flat file

Cleaned output is written to:

s3://nexus-web-frontend1/cleaned-output/<filename>.csv

Screenshot Suggestions:

- Glue job script code
- Sample cleaned output file in S3
- Before vs After comparison of the CSV (raw vs cleaned)

The screenshot shows the AWS Glue Studio console. The left sidebar contains navigation links for AWS Glue, Data Catalog, and Data Integration and ETL. The main panel is titled 'AWS Glue Studio' and features a 'Create job' section with three options: 'Visual ETL', 'Notebook', and 'Script editor'. Below this is an 'Example jobs' section with a 'Create example job' button. The 'Your jobs (1)' section displays a table of existing jobs.

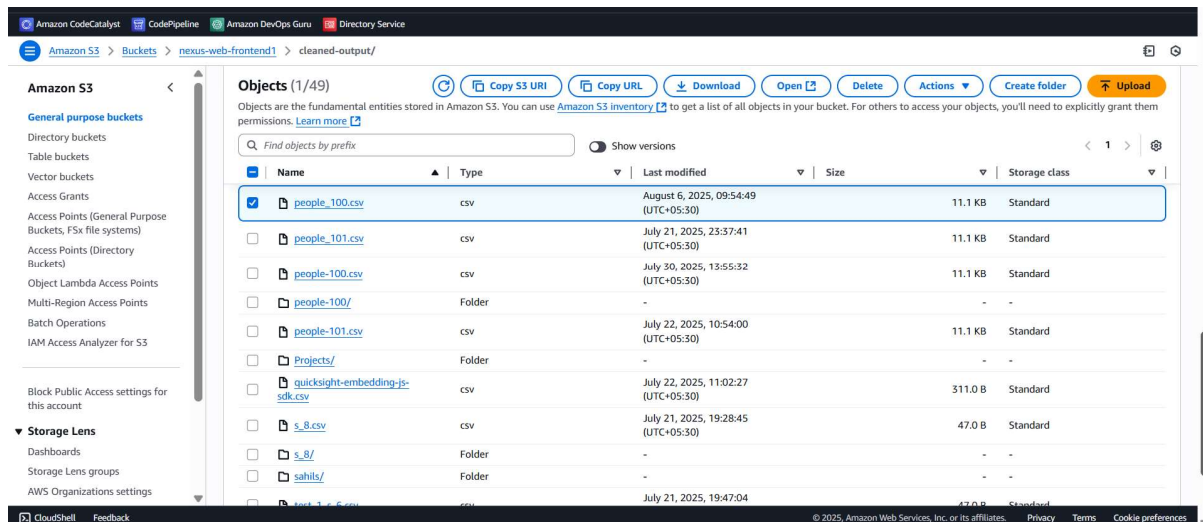
Job name	Type	Created by	Last modified	AWS Glue version
csv-cleaning-job	Glue ETL	Script	8/13/2025, 11:20:46 AM	5.0

The screenshot shows the 'Script' editor for the 'csv-cleaning-job'. The left sidebar is the same as the previous screenshot. The main panel has tabs for 'Script', 'Job details', 'Runs', 'Data quality', 'Schedules', and 'Version Control'. The 'Script' tab is active, showing a Python script for cleaning CSV data. The script includes imports for sys, os, boto3, time, json, and SparkContext, and defines a job function.

```
1 import sys
2 import os
3 import boto3
4 import time
5 import json
6 from awsglue.utils import getResolvedOptions
7 from pyspark.context import SparkContext
8 from awsglue.context import GlueContext
9 from awsglue.job import Job
10 from pyspark.sql.functions import col
11
12 args = getResolvedOptions(sys.argv, ['JOB_NAME', 'S3_INPUT_FILE'])
13 filename = os.path.basename(args['S3_INPUT_FILE']).split('.')[0]
14
15 sc = SparkContext()
16 glueContext = GlueContext(sc)
17 spark = glueContext.spark_session
18 job = Job(glueContext)
19 job.init(args['JOB_NAME'], args)
```

The screenshot shows the Amazon S3 console for the 'nexus-web-frontend1' bucket. The left sidebar contains navigation links for Amazon S3, General purpose buckets, and Storage Lens. The main panel has tabs for 'Objects', 'Properties', 'Permissions', 'Metrics', 'Management', and 'Access Points'. The 'Objects' tab is active, showing a list of objects in the bucket.

Name	Type	Last modified	Size	Storage class
about.html	html	July 22, 2025, 12:59:18 (UTC+05:30)	16.0 KB	Standard
athena-output/	Folder	-	-	-
cleaned-output/	Folder	-	-	-
contact.js	js	July 22, 2025, 12:59:18 (UTC+05:30)	1.0 KB	Standard
dashboard.html	html	July 26, 2025, 13:49:35 (UTC+05:30)	4.8 KB	Standard
Data/	Folder	-	-	-
faq.html	html	July 22, 2025, 12:59:18 (UTC+05:30)	12.1 KB	Standard
features.html	html	July 22, 2025, 12:59:18	6.7 KB	Standard



Step 6: Advanced Cleaning with Glue DataBrew (Optional)

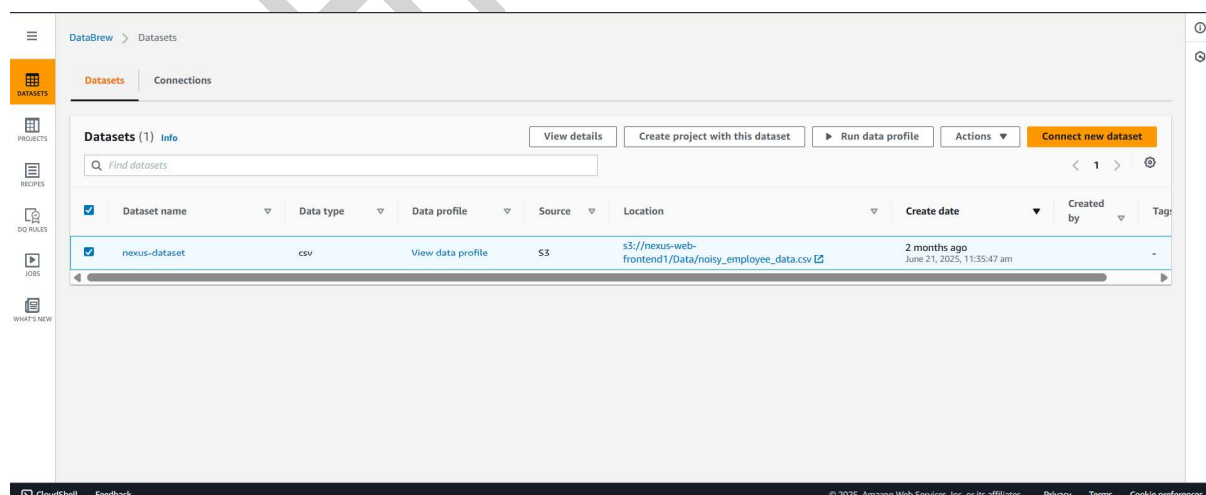
- For more complex or no-code transformations, AWS Glue DataBrew jobs can be triggered.
- It enables:
 - Data profiling
 - Pattern matching
 - Custom transformations

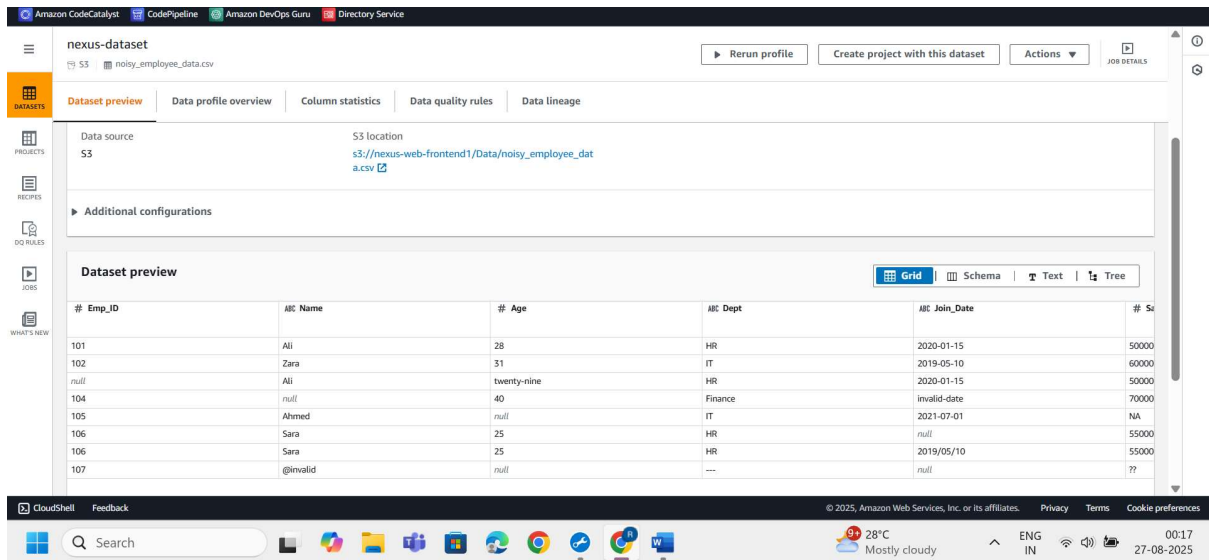
✂ Technologies Used:

- AWS Glue DataBrew

📷 Suggested Screenshots:

- DataBrew project and recipe
- Preview of cleaned results
- Profile summary (data quality insights)





Amazon CodeCatalyst CodePipeline Amazon DevOps Guru Directory Service

nexus-dataset

Dataset preview Data profile overview Column statistics Data quality rules Data lineage

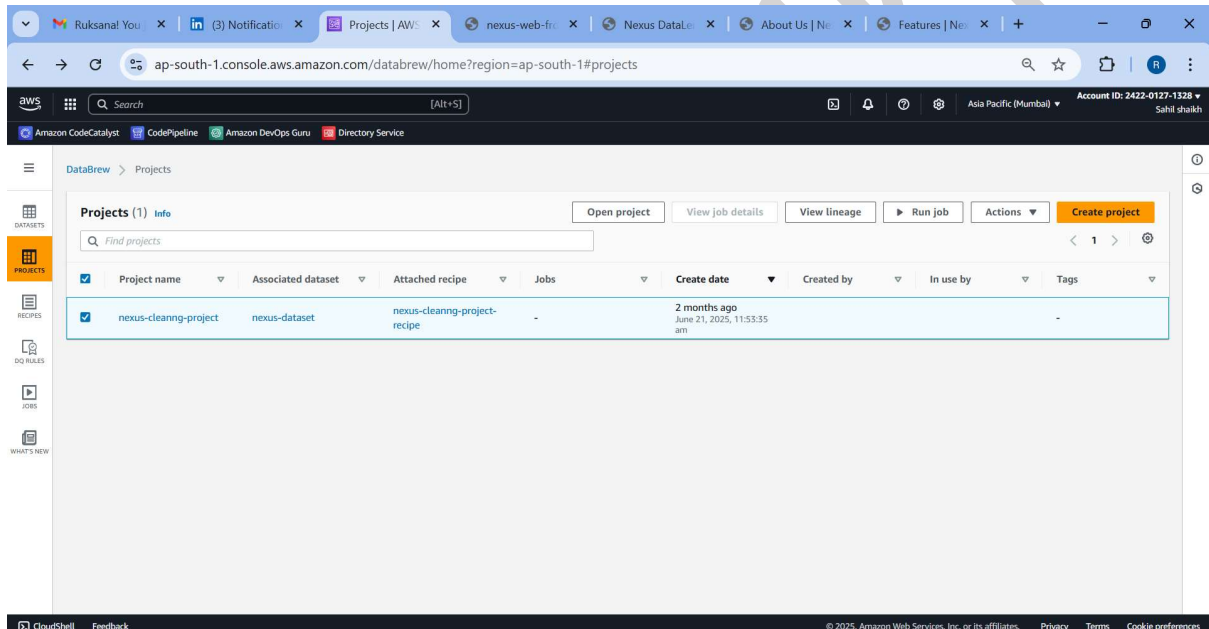
Data source S3 S3 location s3://nexus-web-frontend1/Data/noisy_employee_data.csv

Additional configurations

Dataset preview

# Emp_ID	ABC Name	# Age	ABC Dept	ABC Join_Date	# Salary
101	Ali	28	HR	2020-01-15	50000
102	Zara	31	IT	2019-05-10	60000
null	Ali	twenty-nine	HR	2020-01-15	50000
104	null	40	Finance	invalid-date	70000
105	Ahmed	null	IT	2021-07-01	NA
106	Sara	25	HR	null	55000
106	Sara	25	HR	2019/05/10	55000
107	@invalid	null	---	null	??

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ap-south-1.console.aws.amazon.com/databrew/home?region=ap-south-1#projects

Amazon CodeCatalyst CodePipeline Amazon DevOps Guru Directory Service

DataBrew > Projects

Projects (1) Info

Find projects

Project name	Associated dataset	Attached recipe	Jobs	Create date	Created by	In use by	Tags
nexus-cleaning-project	nexus-dataset	nexus-cleaning-project-recipe	-	2 months ago June 21, 2025, 11:53:35 am			

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The screenshot displays the AWS DataLens console for a project named 'nexus-cleaning-project'. The top navigation bar shows various AWS services like Amazon CodeCatalyst, CodePipeline, Amazon DevOps Guru, and Directory Service. The main workspace area is currently in a 'Provisioning compute' state, indicated by a modal dialog that says 'Your session will be ready soon! Provisioning compute' with a progress bar at 0%. The workspace also shows a 'Dataset: nexus-dataset' and a 'Sample: First n sample (0 rows)'. On the right-hand pane, a 'Recipe (8)' is listed, titled 'nexus-cleaning-project-recipe' (Version 4.0). The recipe includes 8 applied steps: 1. Remove duplicates from Name, 2. Remove duplicates from Emp_ID, 3. Remove duplicates from Age, 4. Remove duplicates from Dept, 5. Remove duplicates from Join_Date, 6. Remove duplicates from Salary, 7. Delete empty rows with missing values in Emp_ID, and 8. Delete empty rows with missing values in Name. The bottom of the screen shows the Windows taskbar with various application icons and system information like temperature (27°C) and time (00:17 on 27-08-2025).

The screenshot shows the AWS Data Catalog console for the 'nexus-cleaning-project' workspace. The interface includes a top navigation bar with the AWS logo, search bar, and account information (Asia Pacific (Mumbai), Account ID: 2422-0127-1328, Sahil shaihk). The main content area displays a data grid with columns: Emp_ID, Name, Age, Dept, Join_Date, and Salary. The grid shows data for three employees: Ali, Zara, and Zara. A 'Recipe (8)' panel on the right lists data cleaning steps: 1. Remove duplicates from Name, 2. Remove duplicates from Emp_ID, 3. Remove duplicates from Age, 4. Remove duplicates from Dept, 5. Remove duplicates from Join_Date, 6. Remove duplicates from Salary, 7. Delete empty rows with missing values, 8. Delete empty rows with missing values in Name. A notification box states 'Your session is ready! You can work with your data now 100%'.

#	Emp_ID	Name	Age	Dept	Join_Date	Salary
101	101	Ali	28	HR	2020-01-15	50000
102	102	Zara	31	IT	2019-05-10	60000

ap-south-1.console.aws.amazon.com/databrew/home?region=ap-south-1#recipes

What is a recipe

Recipes (1) Info

Download as YAML Download as JSON Create job with this recipe Actions Upload recipe

Find recipes Published

Recipe name	Version description	Associated projects	Published date	Published by	Tags
nexus-cleaning-project-recipe Published version 4.0	-	nexus-cleaning-project	2 months ago July 10, 2025, 5:23:13 pm	-	-

ap-south-1.console.aws.amazon.com/databrew/home?region=ap-south-1#recipe-details?recipe=nexus-cleaning-project-recipe&tab=steps

nexus-cleaning-project-recipe

Project: nexus-cleaning-project

Create job with this recipe Actions OPEN PROJECT

Current version
Version 4.0
Published version

Version 3.0
Published version

Version 2.0
Published version

Version 1.0
Published version

Version 0.2
Job run version

Version 0.1
Job run version

Recipe steps

Recipe details

Recipe name: nexus-cleaning-project-recipe

Associated project: nexus-cleaning-project

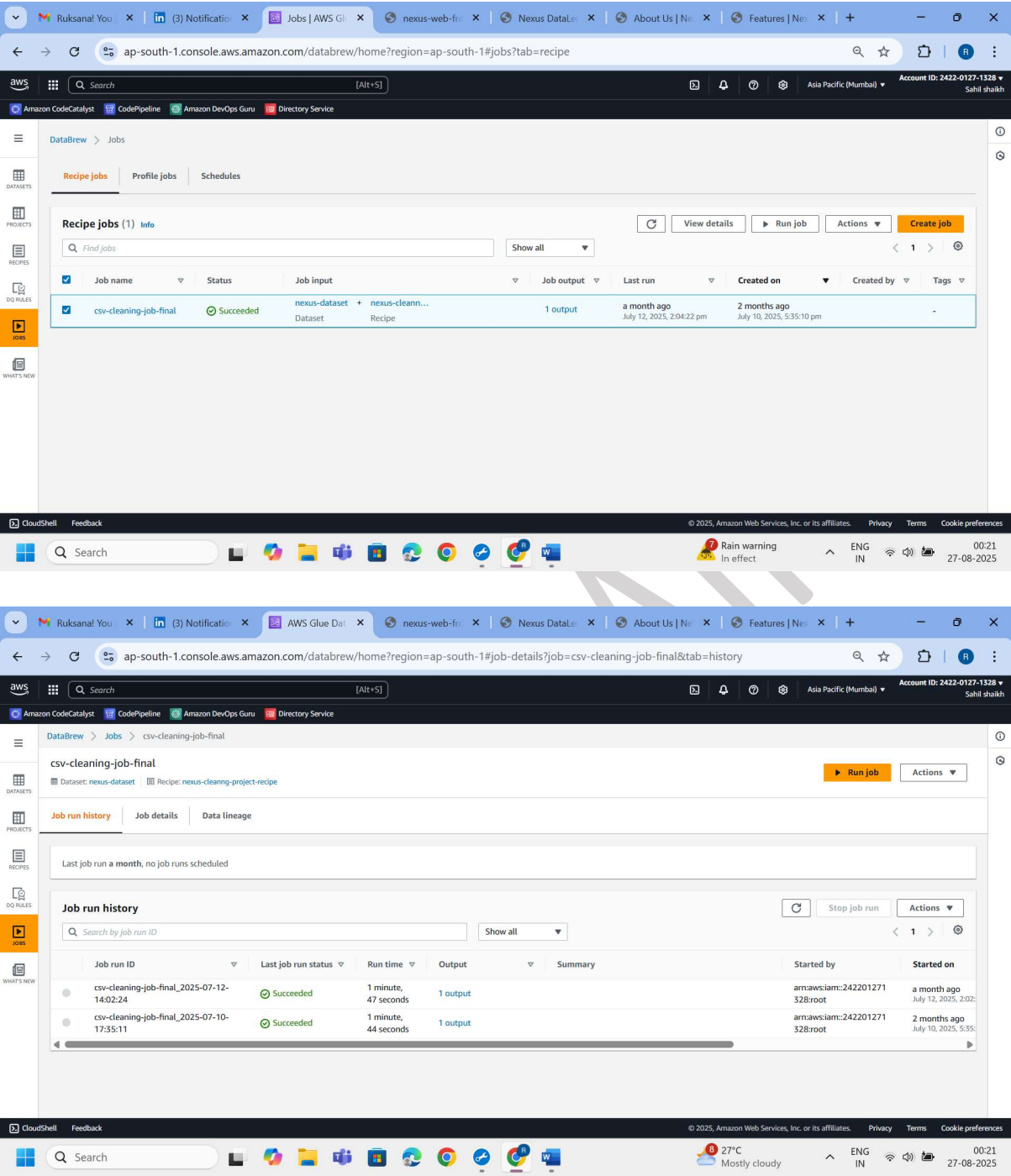
Published date: 2 months ago by arnaws:iam::242201271328:root

Version description: -

Associated jobs: csv-cleaning-job-final

Recipe steps (8)

1. Remove duplicates from Name
2. Remove duplicates from Emp_ID
3. Remove duplicates from Age
4. Remove duplicates from Dept
5. Remove duplicates from Join_Date
6. Remove duplicates from Salary



Step 5: Second Lambda Invocation

Once the Glue Job completes, it **invokes a second Lambda function** to automate the following:

- Trigger a second Glue Crawler for the cleaned data
- Configure Athena dataset
- Automatically generate a QuickSight dashboard using a predefined template

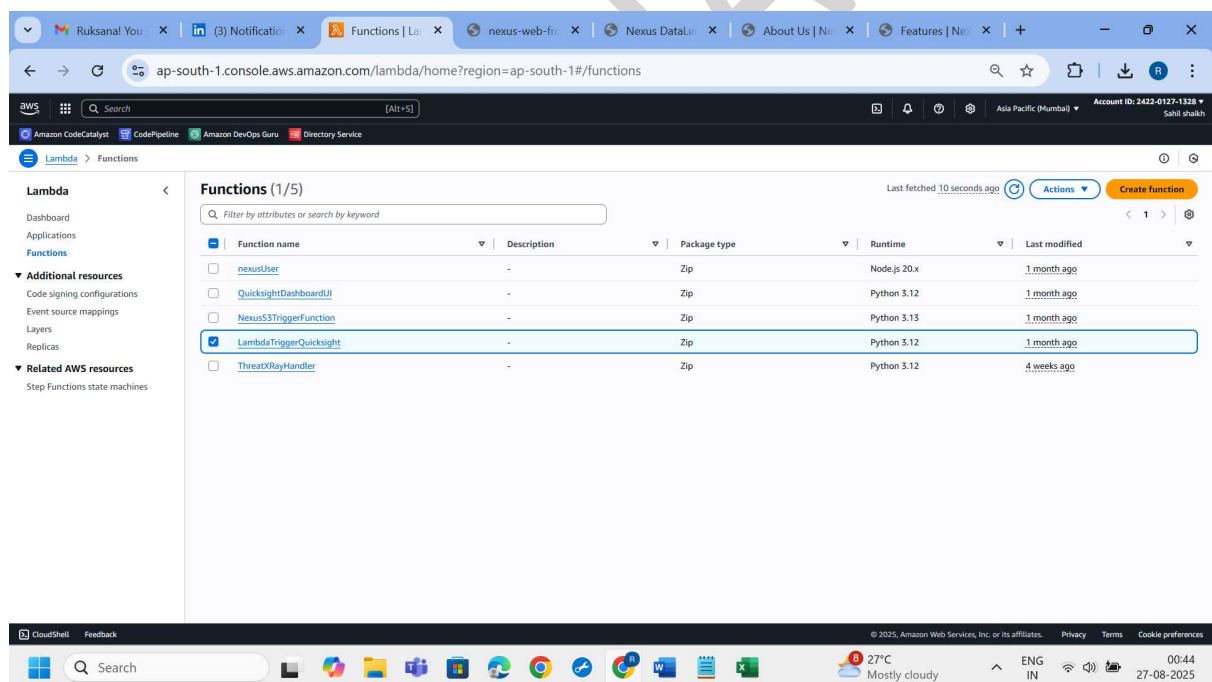
✂ Technologies Used:

- AWS Lambda
- AWS Glue Crawler
- Athena dataset creation via Boto3
- QuickSight automation via SDK



Screenshot Suggestions:

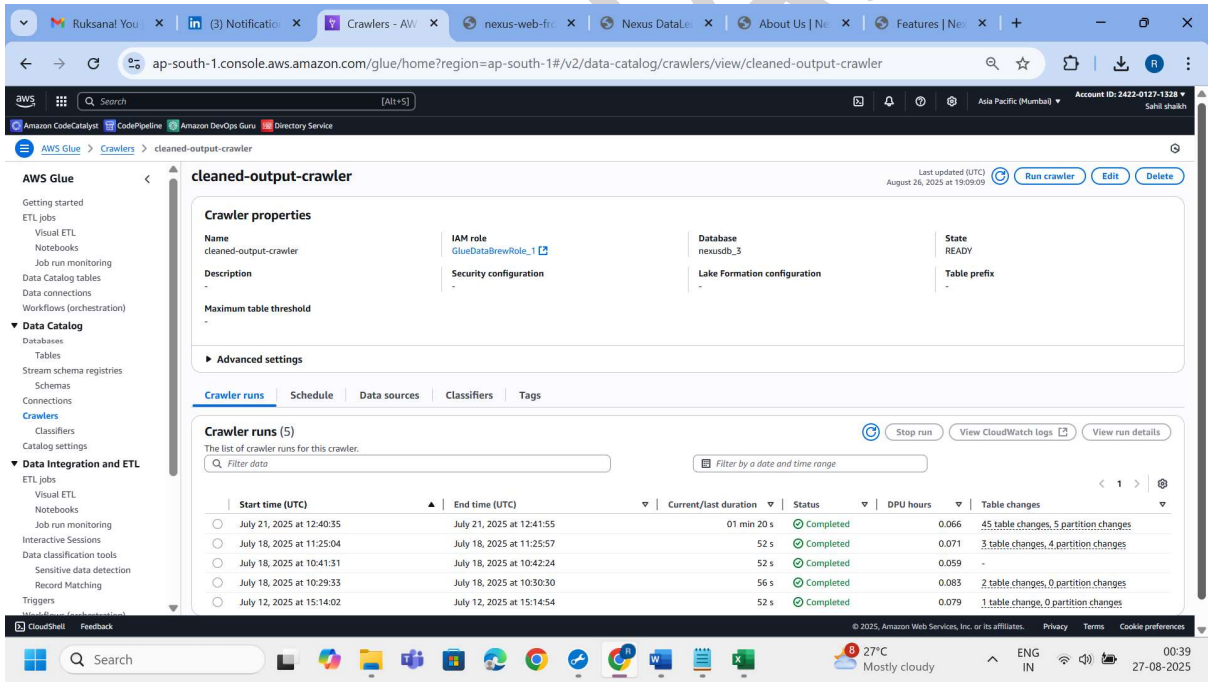
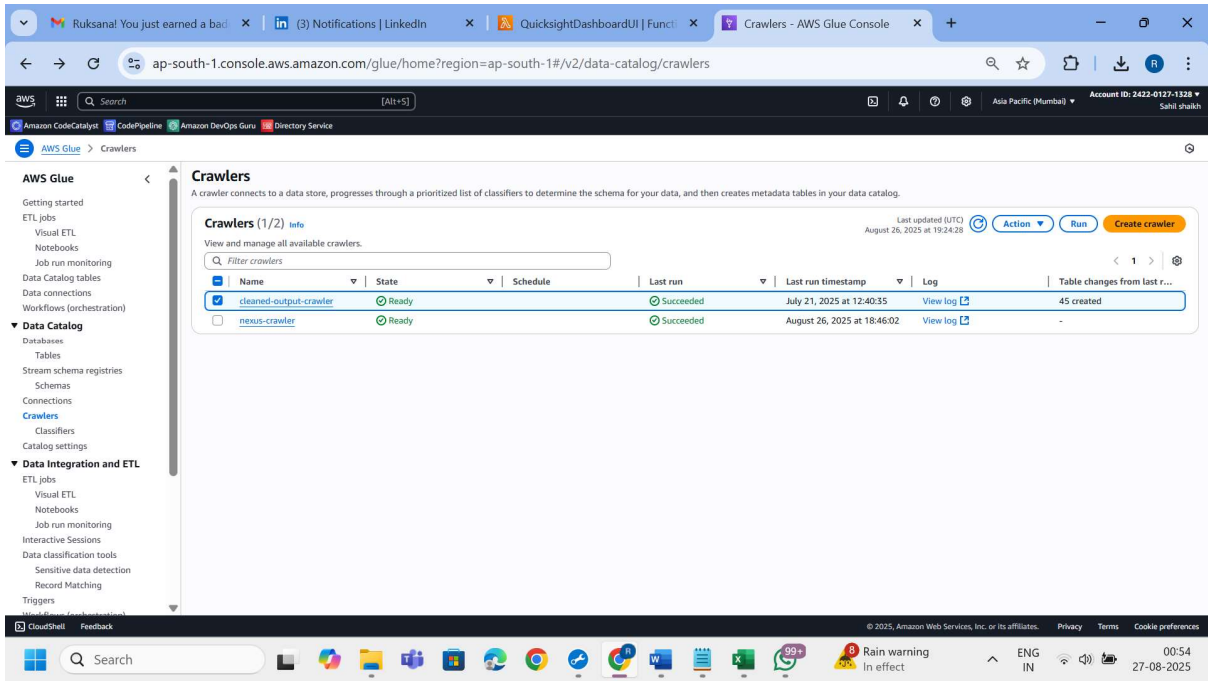
- Lambda code snippet invoking dashboard generation
- Glue Crawler for cleaned output



The screenshot shows the AWS Lambda console for the function 'LambdaTriggerQuicksight'. The 'Function overview' tab is active, displaying a diagram of the function's configuration. It shows triggers for 'EventBridge (CloudWatch Events)' and 'S3'. The 'Description' panel on the right provides details: 'Last modified' 1 month ago, 'Function ARN' arn:aws:lambda:ap-south-1:242201271328:function:LambdaTriggerQuicksight, and 'Function URL' info. The 'Code source' tab is also visible, showing the 'lambda_function.py' file.

The screenshot shows the AWS Lambda console for the function 'LambdaTriggerQuicksight', with the 'Code source' tab active. The 'lambda_function.py' file is open, displaying the following code:

```
1 import boto3
2 import json
3 import os
4 import uuid
5
6 quicksight = boto3.client('quicksight')
7 glue = boto3.client('glue')
8 sts = boto3.client('sts')
9 account_id = sts.get_caller_identity()['Account']
10
11 # Valid Quicksight column types
12 TYPE_MAPPING = {
13     "string": "STRING",
14     "bigint": "INTEGER",
15     "int": "INTEGER",
16     "integer": "INTEGER",
17     "double": "DECIMAL",
18     "float": "DECIMAL",
19     "boolean": "BOOLEAN",
20     "timestamp": "DATETIME",
21     "date": "DATETIME",
22     "binary": "STRING",
23     "char": "STRING",
24     "varchar": "STRING"
25 }
26
27 def map_field_type(qs_col):
28     col_type = qs_col['Type']
29     if col_type in ["INTEGER", "DECIMAL"]:
30         return "NumericalDimensionField"
31     elif col_type == "DATETIME":
```



Step 6: Querying via Athena

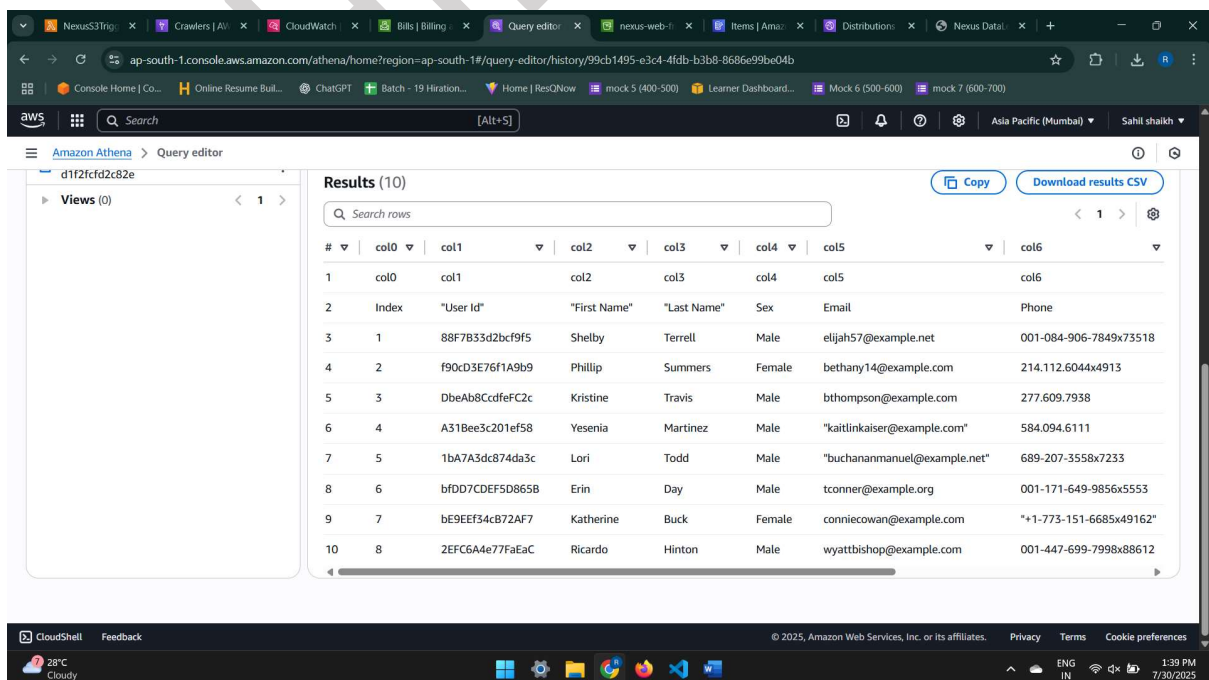
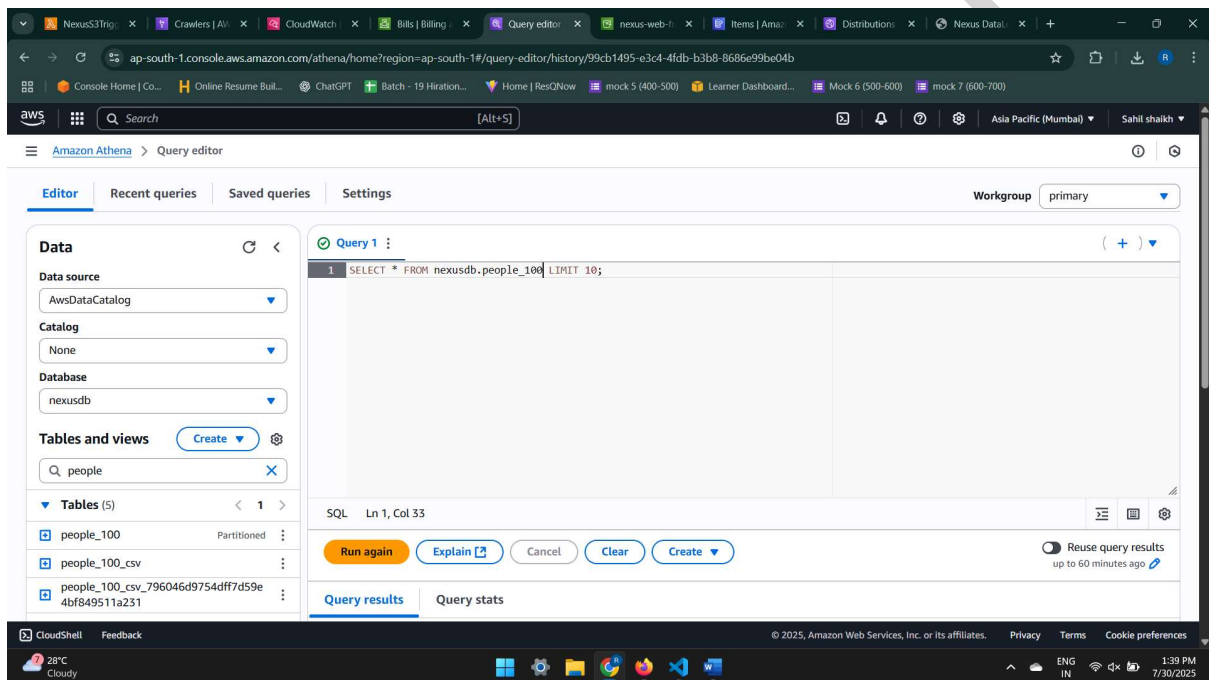
- Athena enables serverless querying using standard SQL.
- Cleaned data in S3 is queried via the Glue Catalog tables.

Example query:

```
SELECT * FROM nexusdb.cleaned_<filename> LIMIT 10;
```

 *Screenshot Suggestions:*

- Athena query editor with sample query
- Preview of query results



Step 7: Data Visualization – QuickSight & Power BI

- In **Amazon QuickSight**, dashboards are generated automatically using Athena datasets.
- In **Microsoft Power BI**, users can connect via the **Athena ODBC driver** for custom reporting.

🔧 Technologies Used:

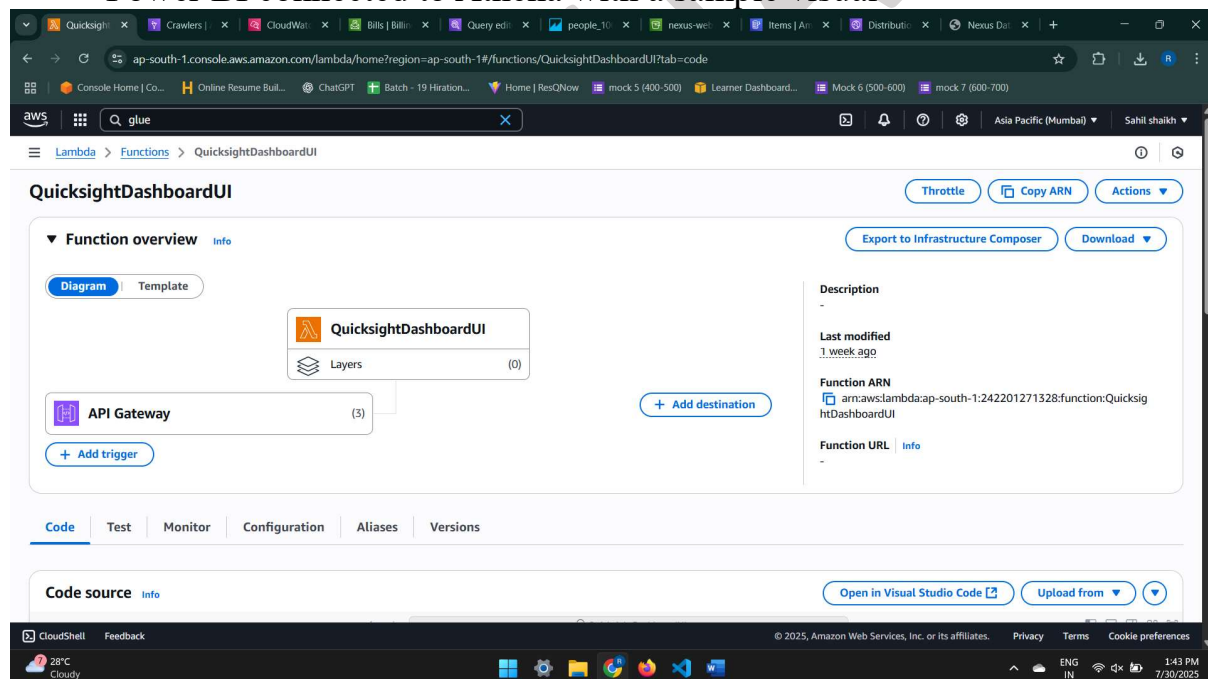
- Amazon QuickSight
- Microsoft Power BI (Optional)
- Athena ODBC Connector

📸 Suggested Screenshots:

- QuickSight dashboards (bar, pie, line)
- Power BI report preview (same dataset)
- ODBC config panel in Power BI

📸 Screenshot Suggestions:

- QuickSight dashboard with graphs/charts
- Power BI connected to Athena with a sample visual



The screenshot displays the Amazon QuickSight interface. At the top, the browser address bar shows the URL: `ap-south-1.quicksight.aws.amazon.com/sn/data-sets/9f6767a5-e4d4-4871-becc-5ee64fdb8cf2/view`. The main header shows the dataset name **people_100** with buttons for **DELETE DATASET**, **EDIT DATASET**, and **USE IN ANALYSIS**.

The **About** section for the dataset includes:

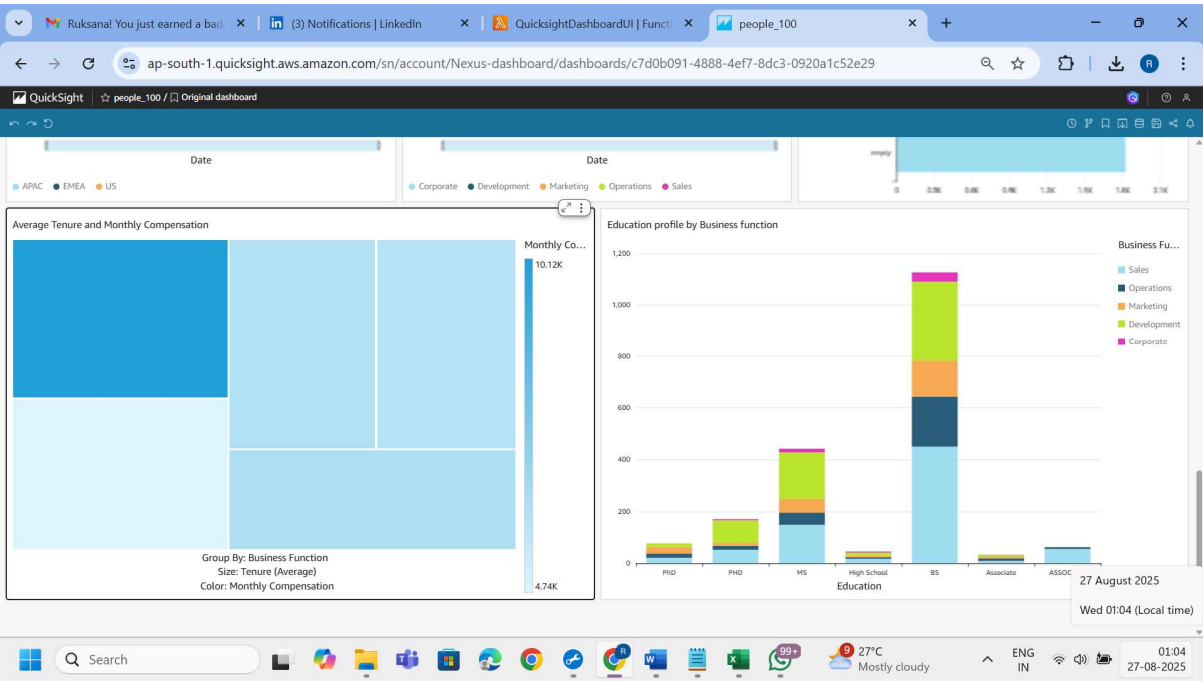
- SPICE** icon and **Size: 34.4 KB**
- REFRESH** status: **Completed**, 102 rows imported (100% success). Last successful refresh: July 21, 2025 at 11:46 PM GMT+5:30.
- ACCESS SETTINGS**: Sharing (Owners: 1, Viewers: 0), Row-level security (No restrictions), Column-level security (No restrictions).
- SCHEMA**: Unique key (Learn more), Disabled.

The **Sources** section lists **newusdb**. The **Usage** section shows **Analyses (1)**, **Dashboards (1)**, and **Datasets (0)**.

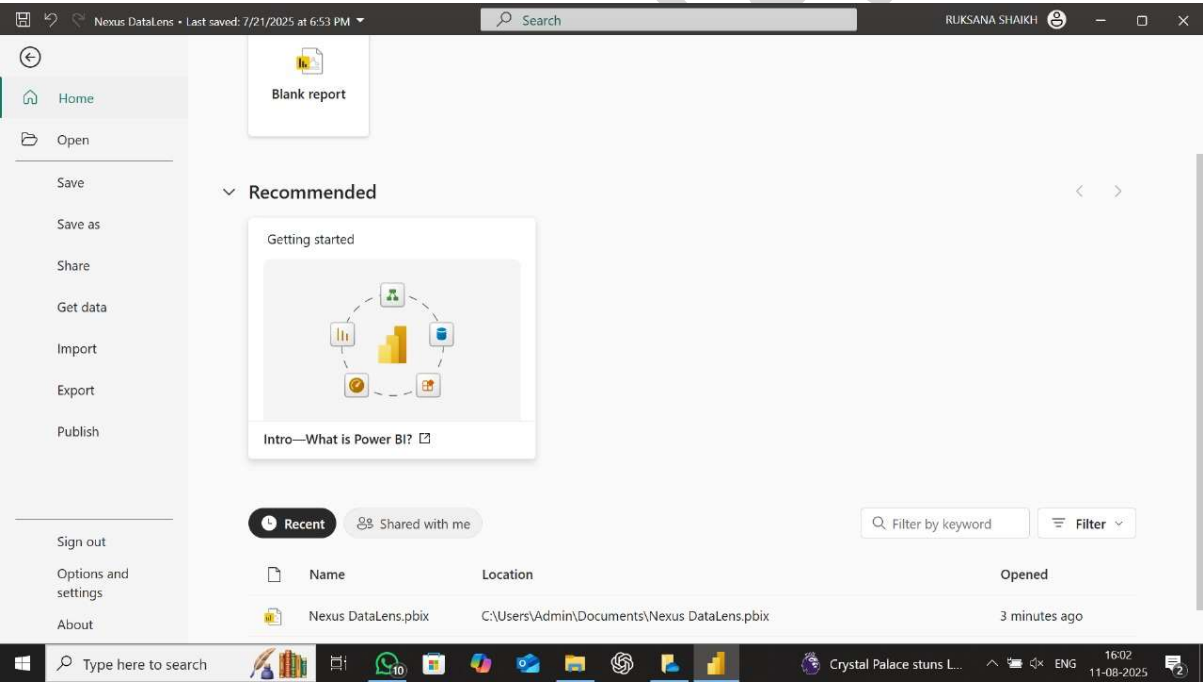
The dashboard view shows four visualizations on a grid:

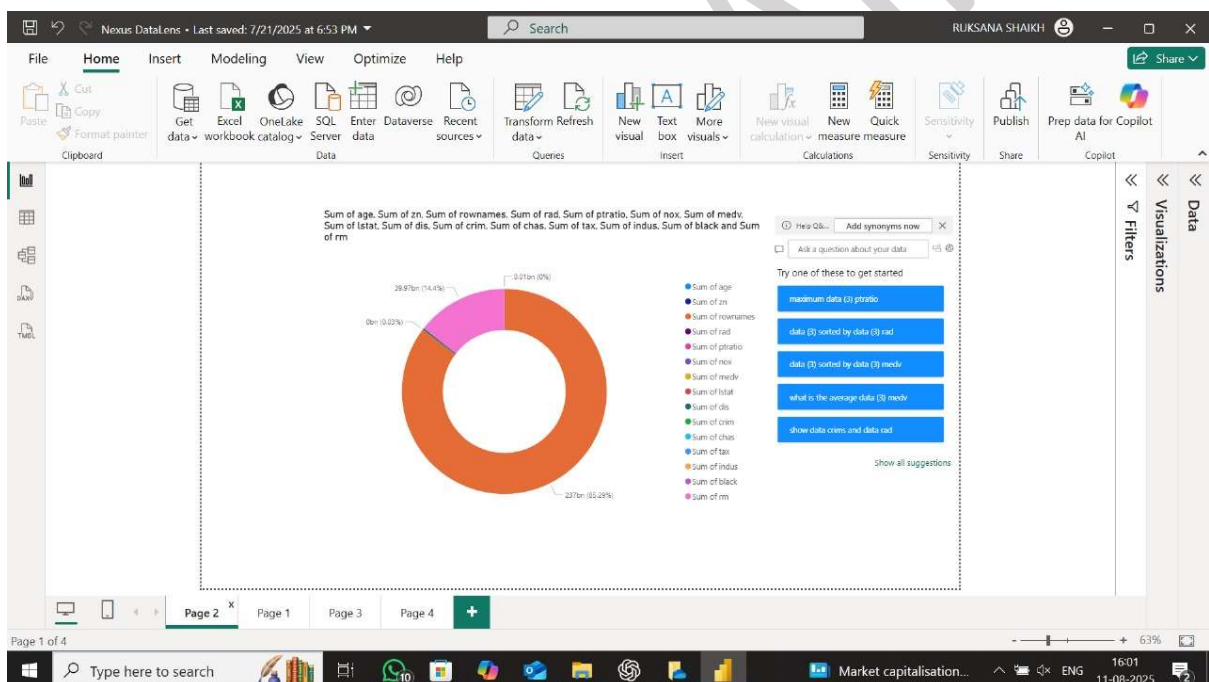
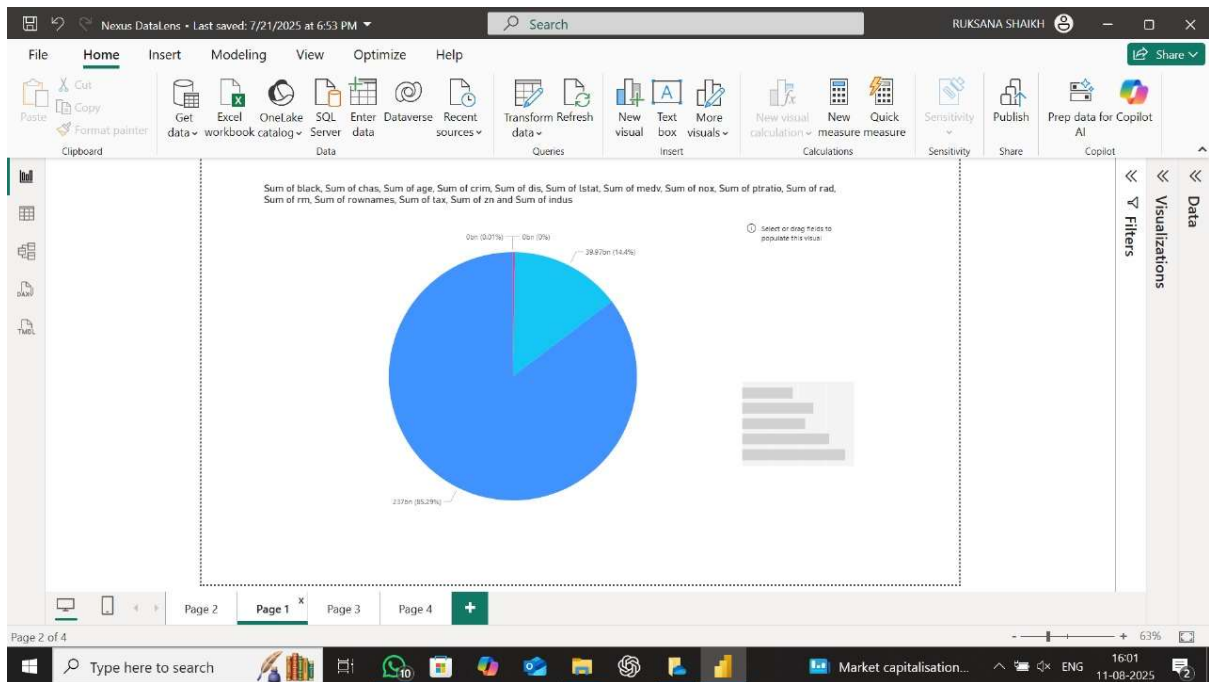
- Count of Col0 by Col0 and Col3**: A horizontal bar chart showing the top 102 in Col0 and bottom 25 in Col3. The y-axis is labeled 'Col 3' and the x-axis is labeled 'Col 0 (Count)'. The chart shows a distribution of values across categories like 'Last Name', 'Andrews', 'Atkinson', 'Bailey', 'Barnes', 'Barnett', and 'Bennett'.
- Count of Col7 by Col5 and Col8**: A pie chart titled 'Accountant' showing the distribution of values across categories like 'clinton78@example.org' and 'clinton78@example.org'.
- Count of Col5 by Col2, Col8, and Col5**: A bar chart showing the distribution of values across categories like 'alexiscantrell@example.org' and 'angelanavarro@example.net'.
- Count of Col7 by Col5 and Col8**: A bar chart showing the distribution of values across categories like 'Broadcast', 'Air broker', 'Sport and', 'Veterinary', 'Systems a...', 'Chiropractor', 'Teacher', 'Neurosurge...', and 'Drilling en...'.

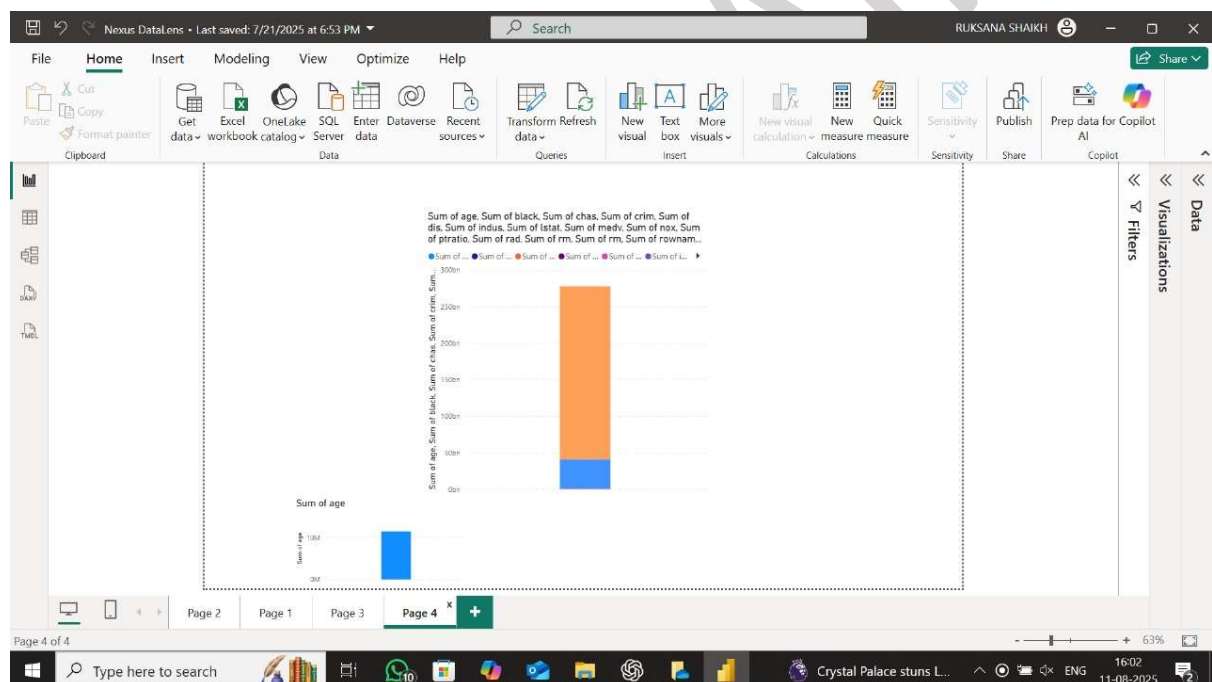
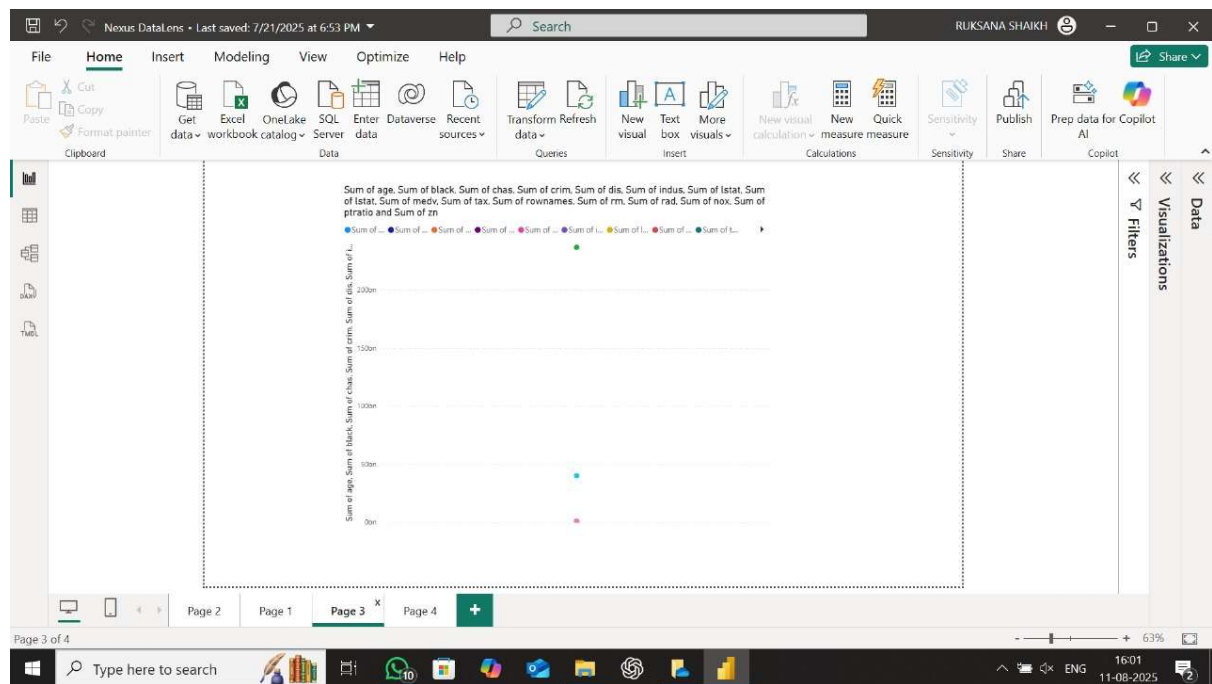
The bottom of the screen shows the Windows taskbar with the date and time: 1:40 PM, 7/30/2025.



POWERBI







6. Data Cleaning Logic

The csv-cleaning-job script implements the following transformations:

- Drop columns and rows that are entirely null
- Filter out invalid or placeholder values
- Rename inconsistent headers to normalized format
- Output a single CSV file for ease of visualization



Screenshot Suggestions:

- Code snippet showing data cleaning logic
- “Before” and “After” preview of CSV structure

7. Final Output and Visualization

The QuickSight dashboard showcases:

- Product-wise sales breakdown
- Monthly or daily trends over time
- Region and category-level filters

Additionally, Power BI integration supports rich business reporting and analysis.



Screenshot Suggestions:

- QuickSight dashboard visuals (bar, line, pie charts)
- Power BI sample visuals for the same dataset

7. Use Cases of Nexus DataLens

The *Nexus DataLens* platform is designed as a scalable, serverless data analytics solution that caters to a variety of real-world business and analytical needs. Its architecture allows organizations to ingest, clean, query, and visualize data with minimal manual effort. Below are key use cases where Nexus DataLens can deliver significant value:

1. Business Intelligence & Reporting Automation

Problem: Businesses often deal with scattered raw data that requires time-consuming preprocessing before analysis.

Solution: Nexus DataLens enables automated transformation of raw CSV files into clean, queryable datasets, and visualizes them through Amazon QuickSight or Power BI dashboards.

Impact: Saves analyst time, enables faster decision-making, and improves data accuracy.

2. Operational Monitoring & KPI Dashboards

Problem: Monitoring KPIs (Key Performance Indicators) across departments can be difficult without centralized visualization.

Solution: Uploads from various teams can be cleaned and visualized automatically into live dashboards using QuickSight or Power BI.

Impact: Enables department heads to monitor sales, operations, customer satisfaction, or finance in real time.

3. Data Democratization for Non-Technical Users

Problem: Non-technical users struggle to query data or generate insights.

Solution: With a simple frontend upload UI, anyone can upload CSV files and get visualized insights without needing to write code.

Impact: Makes data-driven decision-making accessible to marketing, HR, or sales teams.

4. Academic or Research Data Analysis

Problem: Researchers need a quick way to analyze large CSV-based datasets without complex setups.

Solution: Researchers can upload their raw data, clean it, and immediately query or visualize results using built-in analytics tools.

Impact: Accelerates data-driven research and facilitates reproducibility.

5. Client/Partner Data Integration for Enterprises

Problem: Receiving external data from clients or partners often requires preprocessing before use.

Solution: External partners can upload CSVs via the secure frontend; Nexus DataLens automatically prepares it for analysis.

Impact: Speeds up B2B data integration and fosters seamless collaboration.

6. Government or NGO Reporting Dashboards

Problem: Public agencies and NGOs handle data from multiple sources and require transparent reporting.

Solution: Nexus DataLens provides automated processing and visualization, ideal for transparency and compliance.

Impact: Facilitates evidence-based reporting to stakeholders and donors.

7. Startup MVP for Analytics-as-a-Service

Problem: Building a custom analytics backend from scratch is time- and resource-intensive.

Solution: Startups can white-label Nexus DataLens as a backend for providing analytics services to their users.

Impact: Reduces go-to-market time and cost for SaaS products.

8. Internal Tool for ETL Process Validation

Problem: Data engineers often need to test ETL transformations before deployment.

Solution: Engineers can use Nexus DataLens to validate ETL logic on sample CSVs and visualize the output.

Impact: Ensures data quality and reduces post-deployment errors.

8. Key Features & Benefits

Nexus DataLens offers a robust set of features designed to simplify the data analytics lifecycle while ensuring scalability, automation, and enterprise-level security. The following are the key features and associated benefits that distinguish this solution:

1. End-to-End Automation

The entire workflow, from CSV file upload to final dashboard generation, is fully automated. Leveraging AWS Lambda, Glue Crawlers, and Glue Jobs, the system eliminates the need for manual intervention, significantly reducing operational overhead and enabling faster time-to-insight.

2. Intuitive User Experience

The project includes a user-friendly frontend interface hosted on Amazon S3 and protected via AWS Cognito. This intuitive interface allows users to securely upload datasets without technical expertise, making data onboarding seamless and accessible to non-technical stakeholders.

3. Enterprise-Grade Security

Security is built into every layer of the architecture. AWS IAM enforces fine-grained access control, AWS KMS ensures encryption at rest, and AWS CloudWatch enables centralized logging and monitoring. This comprehensive security approach helps maintain compliance and data integrity across all stages of processing.

4. Business Intelligence Flexibility

Nexus DataLens supports dynamic data visualization through both Amazon QuickSight and Power BI. This dual compatibility allows organizations to choose their preferred BI tool, enabling flexibility in reporting, dashboarding, and decision-making processes.

5. Serverless Scalability

Built entirely on serverless AWS services—including Lambda, Glue, Athena, and S3—the solution scales automatically with workload demands. This architecture ensures high availability, cost-efficiency, and consistent performance, regardless of the volume of data or number of concurrent users.

6. Code-Based and No-Code Processing Options

In addition to Spark-based Glue Jobs, the solution integrates AWS Glue DataBrew for users who prefer a visual, no-code environment. This empowers analysts and data stewards to apply complex transformations without writing code, enhancing team productivity and collaboration.

7. Global Accessibility with Optimized Delivery

By integrating Amazon CloudFront as a content delivery network (CDN), the platform ensures fast, secure, and reliable access to the frontend application from any location. This is particularly beneficial for geographically distributed teams and real-time data collaboration.

Here is a professional and formal "**Challenges and Resolutions**" section for your **Nexus DataLens** project documentation:

10. Challenges and Resolutions

Throughout the development of *Nexus DataLens*, several technical and architectural challenges were encountered. Each of these obstacles presented an opportunity to strengthen the system's robustness and reliability. Below is a summary of the key challenges faced and the corresponding solutions implemented.

1. Incorrect S3 Path in Glue Crawler Configuration

Challenge:

Initially, the AWS Glue Crawler was configured to scan an outdated S3 bucket (nexus-web-front), which resulted in the creation of metadata tables pointing to obsolete data locations. This caused downstream services like Athena and QuickSight to display empty or outdated datasets.

Resolution:

The crawler configuration was updated to target the correct S3 path (nexus-web-frontend1/data/). Existing Glue tables linked to the old location were manually deleted from the Data Catalog, and a fresh crawl was initiated to reflect accurate schema metadata in Athena.

2. Frontend Upload Failure Due to S3 Bucket Policy

Challenge:

CSV file uploads from the frontend to S3 were initially unsuccessful due to restrictive CORS policies and insufficient S3 permissions.

Resolution:

The S3 bucket policy was carefully updated to allow secure PUT access from the frontend domain. Cross-Origin Resource Sharing (CORS) rules were configured to allow the appropriate HTTP methods. This enabled seamless file transfers while maintaining security boundaries.

3. Lambda Execution Timeout During Large File Processing

Challenge:

When handling larger datasets, the Lambda function occasionally exceeded its default timeout, disrupting the automated trigger for Glue Crawlers and Jobs.

Resolution:

The function timeout setting was increased within AWS Lambda to accommodate longer processing durations. Logging and exception handling were also enhanced to improve reliability and traceability for future executions.

4. Duplicate Schema Entries in Glue Catalog**Challenge:**

Multiple crawls on the same S3 location led to the creation of duplicate tables in the Glue Catalog, causing confusion in downstream Athena queries and QuickSight reports.

Resolution:

To resolve this, a naming convention and scheduled crawler execution strategy were implemented. Old tables were cleaned up, and table creation settings were restricted to avoid duplication unless schema changes were detected.

5. QuickSight Dataset Not Refreshing Automatically**Challenge:**

After successful data cleaning and schema updates, QuickSight dashboards were not reflecting the latest data due to the absence of automatic dataset refresh.

Resolution:

A manual refresh was triggered initially. Later, this was resolved by integrating QuickSight dataset update logic via Lambda post Glue Job execution. This ensured the dashboard always presents the most recent data.

10. Recommendations for Future Improvements

To further enhance the robustness, usability, and scalability of the Nexus DataLens platform, several strategic improvements are proposed. These recommendations aim to optimize performance, reduce manual overhead, and extend functionality to support enterprise-scale analytics needs.

1. File Validation Prior to Processing

Implementing pre-ingestion validation checks for uploaded files—such as verifying size limits, acceptable formats (e.g., CSV only), and schema conformity—would help prevent processing failures and ensure data consistency.

2. Automated Cleanup of Obsolete Data

Introduce lifecycle management for raw and processed datasets stored in Amazon S3. Automating the deletion of outdated files, logs, and dashboards would not only reduce storage costs but also maintain a clutter-free data environment.

3. Event-Based Notifications

Incorporating Amazon SNS or SES to notify users upon pipeline completion, failure, or anomalies would provide real-time operational transparency. This is particularly useful in multi-user or team environments where timely updates are crucial.

4. Integration with Redshift or RDS

Extending data storage and querying capabilities by integrating with Amazon Redshift or Amazon RDS would enable persistent and structured storage for large-scale data warehousing and advanced analytics.

11. Future Scope

Looking ahead, the Nexus DataLens platform can evolve significantly to support broader use cases and provide deeper insights:

- **Support for Additional File Formats**
Extend compatibility beyond CSV to include formats such as Excel (XLSX) and JSON, thereby enabling a wider range of data sources to be processed seamlessly.
- **Machine Learning-Based Anomaly Detection**
Integrate Amazon SageMaker or similar ML services to automatically detect anomalies and outliers during the data cleaning process, enabling smarter and more proactive data quality assurance.
- **Role-Based Dashboards and Access Controls**
Enhance user experience by enabling personalized dashboards based on roles, departments, or teams. AWS Cognito and IAM policies can be used to implement granular access controls.
- **Comprehensive Notification System**
Expand the notification mechanism to support both email (SES) and SMS-based alerts (SNS), offering flexible communication options tailored to user preferences.

11. Screenshots to Include

 Suggested Visuals (to enhance presentation):

Screenshot

Frontend Upload Page

S3 Buckets

Glue Job Editor

Glue Crawler Config

Athena Query Output

QuickSight Dashboard

Power BI Dashboard

Full Architecture Diagram

Purpose

Show how non-tech users upload CSV files

data/ and cleaned-output/ folders

ETL logic, cleanup process

Scan path and database info

SQL results for verification

Auto-generated visualizations

Manual or semi-automated visual insights

Visual summary of AWS flow

Screenshots

 Include the following:

- Frontend upload UI
- S3 file structure before and after cleaning
- Lambda function execution
- Glue Crawler and Job screens
- Athena query screen
- QuickSight dashboard

12. Conclusion

Nexus DataLens is a scalable, cloud-native solution for automating data ingestion, cleaning, and visualization using AWS services. With seamless integration across Amazon QuickSight and Power BI, it empowers flexible business intelligence. The project showcases how serverless architecture can simplify complex data workflows, offering a reliable and extensible framework for real-time analytics.

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